Survey on Embedding Methods Applied to Ontology Matching

GUILHERME SANTOS SOUSA, MELODI team, Institut de Recherche en Informatique de Toulouse, Toulouse, France
RINALDO JOSE LIMA, Universidade Federal Rural de Pernambuco, Recife, Brazil
CASSIA TROJAHN, MoEx team, University Grenoble Alpes, Saint-Martin-d'Hères, France and Institut de Recherche en Informatique de Toulouse, Toulouse, France

Ontology matching plays a critical role in data integration, knowledge discovery, and ontology merging by establishing correspondences between entities from diverse ontologies. Although traditional methods have proven effective in specific scenarios, they often struggle to capture the semantic depth and contextual nuances of ontology entities, especially when applied to large, heterogeneous, and complex ontologies. Embedding-based methods have gained prominence in recent years. Their effectiveness stems from their ability to encode semantic and contextual information into dense vector representations, which enables more robust similarity computations. This survey provides a comprehensive review of embedding-based techniques applied to ontology matching, with a focus on the schema-level of ontologies. A categorization framework is proposed that identifies the key features of the different proposals and classifies them according to the embedding type, training requirements, and context handling. In addition, this work highlights the challenges and opportunities in embedding-based ontology matching.

CCS Concepts: • **General and reference → Surveys and overviews**; • **Information systems → Ontologies**; • **Computing methodologies → Ontology engineering**; **Learning paradigms**; **Machine learning approaches**.

Additional Key Words and Phrases: Ontology Matching, Embedding, Survey

1 Introduction

Ontology matching is the task of identifying correspondences (usually semantically equivalent entities – classes, properties, and instances) across different ontologies, encompassing both schema-level components (e.g., classes, properties) and instance-level data (e.g., individuals, literals). It is a task recognized as crucial in a range of real applications, such as data integration, knowledge fusion, semantic search, and automated reasoning. Despite the field's maturity, traditional schema-level approaches – based on lexical, structural, or logic-based heuristics – still struggle with challenges such as heterogeneous vocabularies, diverse modeling conventions, and context-dependent semantics. These limitations have motivated a growing interest in embedding techniques embeddings, which offer a data-driven way to capture semantic similarity. In the context of this work, embedding is a learned vector representation of an ontology's entities (such as classes, properties, and individuals). Based on the principles of distributional semantics [31] (that is, the idea that entities occurring in similar contexts tend to have similar meanings), these representations enable semantic comparisons that go beyond surface-level string or structural similarity.

Authors' Contact Information: Guilherme Santos Sousa, MELODI team, Institut de Recherche en Informatique de Toulouse, Toulouse, Occitanie, France; e-mail: Guilherme.Santos-Sousa@irit.fr; Rinaldo Jose Lima, Universidade Federal Rural de Pernambuco, Recife, Brazil; e-mail: rinaldo.jose@ufrpe.br; Cassia Trojahn, MoEx team, University Grenoble Alpes, Saint-Martin-d'Hères, Auvergne-Rhône-Alpes, France and Institut de Recherche en Informatique de Toulouse, Toulouse, France; e-mail: cassia.trojahn-dos-santos@univ-grenoble-alpes.fr.



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Consequently, embeddings offer a powerful alternative for ontology matching. Unlike traditional lexical techniques, embeddings map entities into a continuous vector space where geometric proximity reflects semantic similarity. This allows matchers to capture subtle semantic relationships, such as synonymy, hyponymy, and homonymy, which surface-level string comparisons often miss. Furthermore, these representations can handle polysemy by encoding context-specific information, effectively disambiguating entities that have multiple meanings. This makes embedding-based approaches particularly robust in scenarios involving complex, cross-domain, or low-resource ontologies.

Despite rapid advances in applying embeddings to ontology matching, the field lacks a unified and comprehensive synthesis. While existing surveys address narrow subtopics such as the use of external knowledge sources [91], deep learning techniques, [40] the encoding of ontology entities with embeddings [21], and entity alignment [30, 142]; they do not focus on constructing the embeddings specifically for ontology matching tasks. There remains a gap in understanding how embedding methods are used, what kinds of information they encode, and how effective they are in various matching scenarios, particularly complex ones. This survey addresses the following key research questions: RQ1: *How are embeddings used to represent ontological entities for matching?* RQ2: *What types of embedding strategies have been applied to ontology matching, and how do they differ?* RQ3: *How do existing methods handle the complexity and heterogeneity of real-world ontologies?* RQ4: *What are the limitations of current embedding-based approaches, and what directions should future research take?* To answer these questions, this paper makes the following contributions:

- Comprehensive Survey: provide an up-to-date and systematic review of embedding approaches applied to ontology matching at the schema-level, with a particular focus on embedding-based methods.
- Novel Classification Framework: introduce a new classification model for embedding-based ontology matching approaches, based on three key dimensions: Lexical unit segmentation, Training strategy, and Complexity of information representation (simple vs. complex matching).
- Gap Analysis and Comparison: analyze the strengths and limitations of each matcher, highlighting underexplored areas such as the use of embeddings in complex matching tasks.
- Extension of Prior Work: This survey expands on the work of Sousa et al. [105] by incorporating 81 recent studies and introducing an additional dimension (simple vs. complex matching) to capture task complexity. It reflects the degree of expressiveness supported by embedding-based approaches in the correspondences they generate.
- Future Research Directions: provide a forward-looking discussion on the role of embeddings in ontology matching, including how they can complement external knowledge sources and improve alignment quality in complex scenarios.

The remainder of this paper is organized as follows. Section 2 introduces foundational concepts in ontology matching and embeddings. Section 3 describes the methodology used to identify and analyze relevant works. Section 4 presents the proposed classification model. Section 5 categorizes and describes existing approaches based on the proposed dimensions. Section 6 compares the approaches and discusses key insights. Section 7 outlines promising directions for future research. Finally, Section 8 summarizes the findings and contributions of the survey.

2 Foundational Concepts

2.1 Fundamentals of Ontology Matching

Ontology matching [28] is the process of generating an alignment (A') between ontologies in a set Ω. While Ω can include more than two ontologies in the general case, it typically consists of a pair of ontologies, $\Omega = o_1, o_2$, with alignments expressed as pairs of corresponding entities. Before defining the matching process, the entities of an ontology are introduced.

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Definition 2.1 (Ontology entity). An *ontology entity* $e_i \in o$ is a basic element of the ontology vocabulary. The set of ontology entities E of o is defined as:

$$E(o) = C(o) \cup OP(o) \cup DP(o) \cup I(o)$$

where $C(o)$ denotes the set of classes, $OP(o)$ denotes the set of object properties, $DP(o)$ denotes the set of data properties, $I(o)$ denotes the set of individuals (instances). Thus, an ontology entity e_i satisfies:

$$e_i \in \{\text{class, object property, data property, individual}\}.$$

Each entity is identified by a unique identifier (e.g., an IRI) and may be associated with lexical annotations and formal axioms defining its semantics.

Definition 2.2 (Matching process). The matching process can be defined as a function f that takes a pair of ontologies to be matched (o_1 and o_2), an optional input alignment A, a set of parameters p, and a set of resources r (e.g., background knowledge), and returns an alignment A' between o_1 and o_2:

$$A' = f(o_1, o_2, A, p, r)$$

Each of the elements featured in this definition has specific characteristics that influence the difficulty of the task.

Definition 2.3 (Alignment). An alignment $A_{o_1 \to o_2}$ (or A') is a set of correspondences $c_1, c_2, ..., c_n$. $A_{o_1 \to o_2}$ is directional between a source ontology o_1 and a target ontology o_2.

A correspondence expresses a relation r between ontology entities of o_1 and o_2. Here, the ontology entities are *members* of the correspondence:

Definition 2.4 (Correspondence). A correspondence expresses a relation r (equivalence ($\equiv$), subsumption ($\sqsupseteq$, $\sqsubseteq$)) between two members e_1 and e_2, and n expresses the level of confidence [0..1]. A member can be a single ontology entity of respectively o_1 and o_2, or a more complex construction which is composed of some entities using constructors or transformation functions (as in the examples in the following). From that, two types of correspondences can be considered depending on the type of their members [116]:

- a correspondence is **simple** if both e_1 and e_2 are single entities (represented as IRIs);
- a correspondence is **complex** if at least one of e_1 or e_2 involves a constructor or a transformation function.

The correspondence $\langle e_1, e_2, r \rangle$ asserts that the relation r holds between the ontology entities e_1 and e_2. One has to distinguish between more and less rigorous interpretations of relations. The equivalence relation, for example, can be interpreted as logical equivalence or, more informally, as a high level of similarity. Consider the fragment of the ontologies *ekaw*[1] and *cmt*[2] in Figures 1 and 2, respectively. The format used to represent the ontologies is described in [108]. The following correspondences can be established between these two ontologies:

- *ekaw:Paper,cmt:Paper,*$\equiv$ is a simple correspondence (both members are single entities);
- *ekaw:AcceptedPaper,* $\exists$*cmt:hasDecision.cmt:Acceptance,*$\equiv$) is a complex correspondence with constructors (e_2 involves a constructor). As shown in Figure 2, e_2 corresponds to a fragment of the ontology.

The members of the correspondence can be a simple expression, noted *s*, or a complex expression, noted *c*. A simple correspondence is always *(1:1)* whereas a complex correspondence can be *(1:n), (m:1)* or *(m:n)*. The *(1:1), (1:n), (m:1), (m:n)* notations have been used for the same purpose in the literature [94, 144] (*1* for *s* and *m* or *n* for *c*). However, they can be misinterpreted as the alignment arity or multiplicity.

[1] http://oaei.ontologymatching.org/2019/conference/data/ekaw.owl
[2] http://oaei.ontologymatching.org/2019/conference/data/cmt.owl

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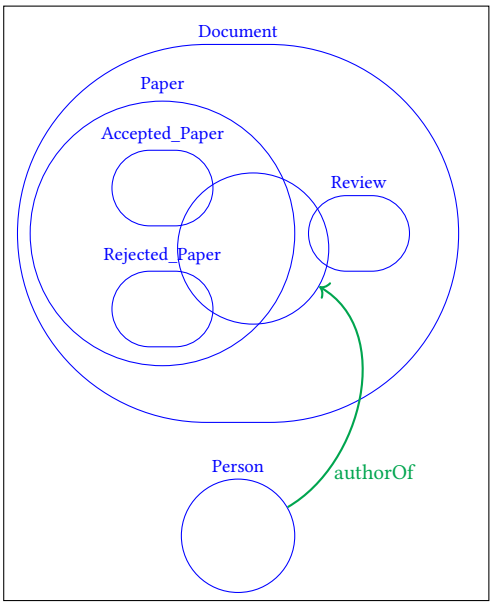


Fig. 1. Fragment of *ekaw* ontology (from [117]).

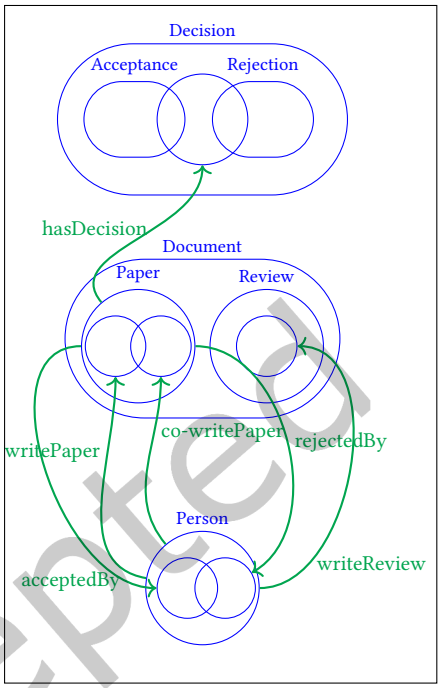


Fig. 2. Fragment of *cmt* ontology (from [117]).

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2.2 Embeddings for Ontologies

An essential step in matching approaches is measuring the *similarity* between ontology entities. Most approaches rely on a lexical step that explores the terminological layer of ontologies (e.g., labels, comments, naming conventions). However, the limited representational capacity of lexical similarity metrics has motivated the integration of alternative techniques. One such technique is the use of embeddings, multidimensional numerical vectors that capture the semantics of various data types, including text, images, symbolic knowledge, and graph structures [141]. These representations can be generated by machine learning models trained on large textual corpora or derived from the structural features of symbolic systems, such as ontologies and knowledge graphs (KGs) [5].

Embeddings enable ontology matching by projecting entities from heterogeneous sources (lexical or structural) into a shared vector space, where semantic similarity can be assessed through geometric proximity, such as the angle between vectors [105]. This representation captures subtle semantic relations, including synonymy, hyponymy, and homonyms, that traditional lexical techniques often overlook [95]. For instance, the embedding vectors for the classes "Paper" and "Contribution" from different ontologies may exhibit high similarity even in the absence of lexical overlap, facilitating the detection of conceptually related but non-identical correspondences, as illustrated in Figure 3. Moreover, embeddings can address polysemy by learning multiple sense-specific representations and selecting the appropriate one based on contextual cues [9],. This capability helps disambiguate entity meanings, thereby improving the robustness and accuracy of ontology matching systems.

The generic embedding-based matchers can be seen as a pipeline of first encoding the ontology elements into embedding vectors and then using these vectors to compare the similarity between entities, as illustrated in Figure 4. The similarity between two entity embeddings is typically measured using the cosine similarity metric, and if this similarity exceeds a predefined threshold, the entities are added to the final alignment set. An ontology

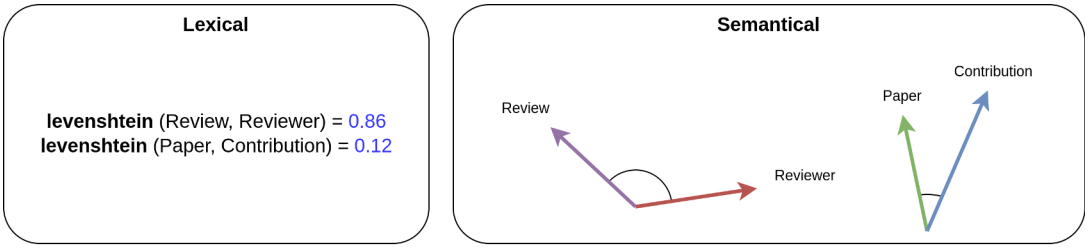


Fig. 3. Embeddings similarity is measured as the angle between two vectors.

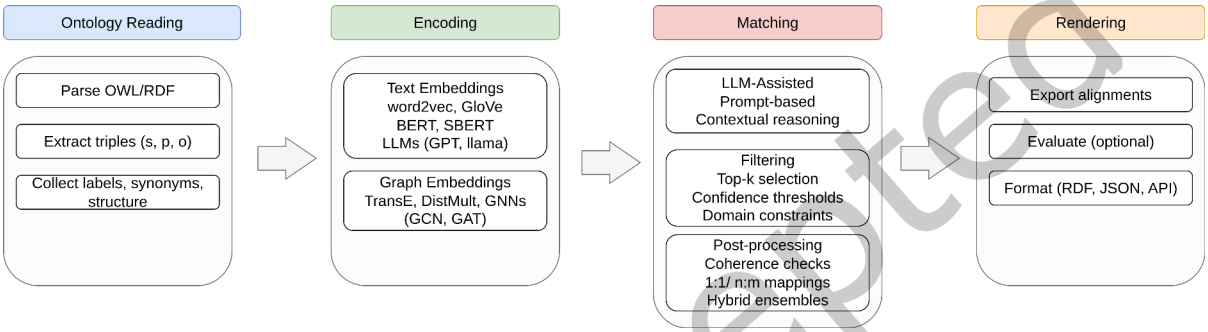


Fig. 4. Overview of the entity alignment process. Entities from two KGs are encoded into a common vector space, after which a similarity metric is applied to the resulting embeddings to determine correspondences between entities.

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matching pipeline can be formally defined as the process designed to identify equivalence correspondences between entities from two distinct ontologies, O1 and O2, given as input. Such a pipeline usually consists of the following steps:

Step 1: Embedding Generation. This initial stage involves creating vector representations for the entities within both ontologies. It can be performed through many techniques, such as geometric modeling, sequence modeling, or graph propagation, and may incorporate both the formal semantics and the textual literals (e.g., names, definitions) associated with the ontology entities.

Step 2: Mapping Discovery. The generated embeddings are then employed to identify potential correspondences. The literature describes many approaches: (i) directly calculating the distance or similarity between the embedding vectors of two entities; (ii) training a supervised classification model that uses the entity embeddings as input features to predict a mapping; and (iii) jointly learning task-specific embeddings with a matching model for optimized performance.

2.2.1 Text Based. Generating ontology embeddings requires accounting for both structural properties and textual metadata. This textual information, primarily entity labels and comments, provides critical natural language descriptions. To encode these elements, common techniques leverage pre-trained word vectors and language models such as Word2Vec [80], GloVe [88], fastText [13], and BERT [24], which offer robust representations derived from large-scale corpora.

Unlike static vectors (Word2Vec, GloVe, and fastText) that assign a single, fixed embedding to each vocabulary term, contextual models such as ELMo [89], BERT, and SentenceBERT [96] generate dynamic representations. While static models fail to distinguish polysemous words, assigning the same vector to "bank" regardless of

meaning, transformer-based architectures leverage multiple neural layers to produce distinct embeddings based on surrounding tokens. This contextual sensitivity is critical when incorporating hierarchical or relational metadata into entity embeddings. For multi-word sequences, representations are typically condensed into a single vector using aggregation strategies such as mean pooling, the [CLS] token in BERT, or the final hidden state in LSTM networks.

An additional distinction among these language models lies in their tokenization process. In Word2Vec, each word corresponds to a single embedding; for example, "Computer" and "Computers" are treated as distinct tokens with their unique embeddings. Differently, models like BERT use subword tokenization with methods such as WordPiece. In this scheme, "Computer" might be split into pieces like "Comp" and "uter", while "Computers" could be segmented into "Comp", "uter", and "s", with each piece having a corresponding embedding. The primary advantage of this subword approach is its ability to handle out-of-vocabulary terms. While Word2Vec cannot generate a representation for a word not present in its vocabulary, WordPiece can construct one by combining the embeddings of its known subword units. This enhances the generalization of matching systems across different domains without the need for retraining.

2.2.2 Graph Based. Along with textual information, in ontology matching, structural embeddings are desired since they can encode more information into entity embeddings to improve matching performance and can deal with the homonym problem to reject false positive correspondences. Diverse techniques for encoding graph structures exist. One of the first approaches is RDF2Vec [97]. In RDF2Vec, the ontology graph is traversed by random walks, generating a document of sentences that are later used by an embedding model to generate the entity embeddings. The original implementation of RDF2Vec splits the sentences in each entity and the predicates connecting them, and uses Word2Vec to generate the final embeddings. OWL2Vec [19] was designed to address the challenges of embedding ontologies written in the OWL language, which provides constructs such as intersectionOf and unionOf for representing complex entities. In OWL2Vec, a set of conversion rules to translate OWL constructs into RDF is used. A similar random walk strategy is applied; however, multiple documents are generated containing lexical, structural, and merged information. This method also generates embeddings for individual entities as well as individual words. Since no labeled data is needed to train these models, they can learn embeddings in an unsupervised way.

Another approach for learning embeddings from graph structures involves translational embeddings [55]. employed in models such as TransE [14], ConvE [23], or RotatE [110]. These methods associate each class and property in an ontology with a randomly initialized embedding vector. The training process is iterative: for a given valid triple $t = (\vec{s}, \vec{p}, \vec{o})$, The learning objective is to minimize the distance between the composite vector $\vec{s} + \vec{p}$ and the target vector $\vec{o}$. Simultaneously, the process maximizes this distance for corrupted triples, which are generated by randomly altering one of the elements in a valid triple. This modeling framework can capture certain hierarchical compositions. For example, the property "grandfather" could be modeled as an embedding vector $\vec{p} = 2\vec{f}$, which is a multiple of the vector $\vec{f}$ for the "father" property. More advanced variations of this model encode embeddings in different spaces or manifolds to capture more complex property patterns [110]. For instance, encoding the properties *writePaper* and *hasRelatedPaper* in a circular space such that they have an angle of 180° degrees between them, a predicate like *inverse*, defined as a rotation of 180° degrees, can transform the embedding of *writePaper* into *hasRelationPaper* by adding *inverse* to it.

Graph Neural Networks (GNNs) [132], encompassing convolutional, message-passing, and attention-based variants, generate node embeddings by iteratively aggregating neighbor information. For matching tasks, these networks refine entity embeddings across multiple layers (see Figure 5). Typically, the embedding of a target entity E_1 is computed by aggregating neighbor embeddings (e.g., E_2 and E_3) and processing the result through a trainable update function. Beyond basic mean or max pooling, property-aware architectures like Relational

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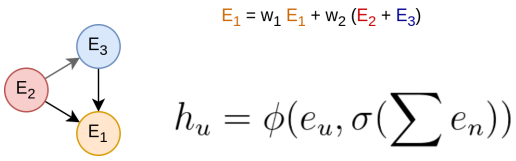


Fig. 5. Computation of E_1 embeddings through neighborhood aggregation using the summation of near embeddings.

Graph Convolutional Networks (R-GCN) [100] explicitly encode relationship types during aggregation, providing richer contextual depth to the final entity representations.

Overall, embedding-based ontology matching provides a powerful and flexible approach for capturing the meaning and relationships of ontology elements and can help overcome the limitations of traditional matching techniques. However, there are also challenges in using embeddings, such as the need for large amounts of training data and the choice of embedding model. The following sections review the state of the art in embedding-based ontology matching techniques and their applications.

3 Survey Method

To identify recent research on embedding-based similarity in ontology matching, this study conducted a literature search across key computer science and artificial intelligence databases. We selected sources for their relevance to OM and machine learning advances, applying keyword-based queries (e.g., ontology matching, embeddings, and similarity calculation) to Google Scholar, Science Direct, ACM, ACL, OAEI, Semantic Scholar, IEEE, arXiv, MDPI, Springer, and HAL. This search spans publications from January 2014 up to July 2025. Detailed search criteria and results per database are summarized in Table 1. Note that keywords were adjusted to accommodate the specific query formats and mechanisms of each search engine.

Base	Link	Keywords	Total	Considered
google scholar	https://scholar.google.com/	ontology embedding matching OR alignment	17,200	100
science direct	https://www.sciencedirect.com/search	ontology embedding matching alignment	1,175	200
acm	https://dl.acm.org/search/advanced	[All: ontology] AND [[All: matching] OR [All: alignment]] AND [All: embedding] AND [E-Publication Date: (01/01/2014 TO *)]	2,639	150
acl	https://aclanthology.org/	ontology AND (match OR align)	40	40
oaei	http://om2023.ontologymatching.org/	*	287	287
semantic scholar	https://www.semanticscholar.org/	ontology embedding matching alignment	969	400
ieee	https://ieeexplore.ieee.org/search/advanced	("All Metadata":ontology) AND ("All Metadata":matching OR "All Metadata":alignment) AND ("All Metadata":model OR "All Metadata":embedding OR "All Metadata":learning)	476	476
arxiv	https://arxiv.org/search/advanced	Query: order: -announced_date_first; size: 200; date_range: from 2014-01-01 ; include_cross_list: True; terms: AND all=ontology; AND all=matching OR alignment; AND all=model OR embedding OR learning	124	124
mdpi	https://www.mdpi.com/search?	ontology matching or ontology alignment	169	169

Table 1. Literature databases and corresponding search queries used in this review.

Following the initial search, the retrieved papers were manually filtered according to three primary selection criteria: accessibility, relevance, and methodology. Publications were excluded if they were inaccessible behind

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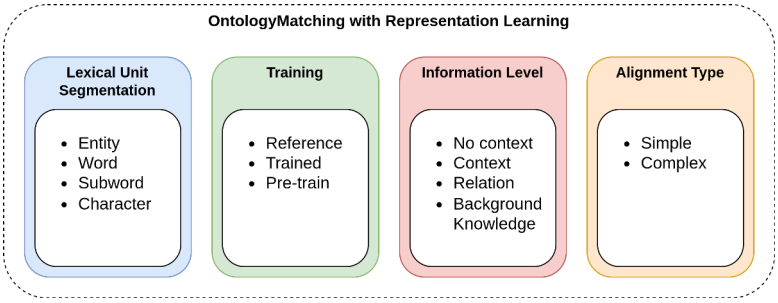


Fig. 6. Overview of the proposed categorization framework.

a paywall, were not directly related to schema-level matching, or did not employ embeddings for similarity calculation. This filtering process was designed to ensure that the selected papers are closely aligned with the survey's research questions and accurately represent recent developments in embedding-based ontology matching. Building upon [105], which analysed 23 papers, this survey incorporates 80 additional publications. The resulting corpus of 103 papers forms the basis of this analysis, offering insights into the effectiveness of embedding-based methods for ontology matching and identifying directions for future research.

4 A Classification Framework for Embedding-Based Ontology Matching

This survey classifies matching systems based on their embedding-based similarity computation, terminological feature handling, and training procedures. Building upon the framework in [105], we extend the "Lexical Unit Segmentation" category to include subword segmentation and introduce a new "Alignment Type" category to distinguish between simple and complex matching. As shown in Figure 6, our framework organizes systems along four non-mutually exclusive dimensions: Lexical Unit Segmentation, Training, Information Level, and Alignment Type. This categorization describes the information aggregated during entity embedding generation. Because these categories overlap, a single model may occupy multiple classifications. For instance, a system using word-level segmentation while incorporating external data like Wikipedia would be classified under both Word-level segmentation and Background Knowledge information levels.

4.1 Lexical Unit Segmentation

This category covers the various levels at which terminological data is segmented in matching systems. The chosen segmentation level significantly impacts a matcher's generalization capabilities and its performance on the out-of-vocabulary (OOV) problem [51]. This classification framework extends the segmentation proposed in [125] by adding the entity level, and is composed of four sub-categories (Figure 6), which are explained in the subsequent sections. An example is given in Figure 7.

Entity-level segmentation generates a unique representation for each entity by treating its textual label as an indivisible unit. Alternatively, a unique embedding can be assigned directly to an entity's identifier, bypassing textual content entirely, a method exemplified by RDF2Vec [97]. In this context, an entity's entire terminological label constitutes a single "sentence" or atomic token. For example, phrases like "accepted paper" and "reviewed book" would each be assigned a single, unique embedding. A primary limitation of this approach is its inability to represent out-of-vocabulary entities; a model cannot generate an embedding for "accepted book" if that specific sequence was absent during training, regardless of whether "accepted" and "book" were seen individually. However, this method is computationally efficient, as the absence of aggregation or composition functions allows for smaller, less complex processing models.

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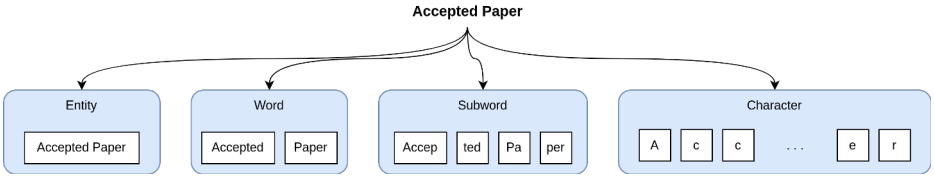


Fig. 7. Example of the label "Accepted Paper" split in each category. The tokenization shown here has illustrative purposes only. Actual implementation can give different results.

Entity-level segmentation generates a unique representation for each entity by treating its textual label as an indivisible unit. Alternatively, a unique embedding can be assigned directly to an entity's identifier, bypassing textual content entirely, a method exemplified by RDF2Vec [97]. In this context, an entity's entire terminological label constitutes a single "sentence" or atomic token. For example, phrases like "accepted paper" and "reviewed book" would each be assigned a single, unique embedding. A primary limitation of this approach is its inability to represent out-of-vocabulary entities; a model cannot generate an embedding for "accepted book" if that specific sequence was absent during training, regardless of whether "accepted" and "book" were seen individually. However, this method is computationally efficient, as the absence of aggregation or composition functions allows for smaller, less complex processing models.

Word-level segmentation generates a distinct representation for each word within an entity's terminological layer. The final entity representation is then computed by applying an aggregation function (e.g., sum or average) to combine these individual word embeddings into a single vector. This is the standard approach adopted by classic word embedding models such as GloVe [88].

For example, under a word-level segmentation scheme, the entity "accepted paper" is tokenized into its constituent words, "accepted" and "paper." After retrieving the individual embedding for each word, an aggregation function is applied to compute a single vector representation for the complete entity. Another aggregation strategy can use the hidden state of a neural network; for instance, the final output of a Long Short-Term Memory (LSTM) network can serve as the composite embedding. This type of segmentation offers better generalization because the model can construct a meaningful representation for an unseen entity, such as "accepted book," as long as the embeddings for its constituent words ("accepted" and "book") are known.

Subword-level segmentation divides text into units smaller than a whole word, such as word fragments or n-grams. This technique is a standard practice in many modern language models. For example, the WordPiece tokenizer, used by BERT, might split the word "accepted" into "accep" and "ted." The vocabulary for such tokenizers is learned from a large text corpus in a preliminary training step. Other common subword segmentation schemes include Byte-Pair Encoding (BPE) [74] and n-gram-based methods.

Character-level segmentation, the most granular approach, operates on individual characters within the terminological layer. Its primary advantage is universality; by utilizing a character-based alphabet, it eliminates the out-of-vocabulary problem and maintains a vocabulary orders of magnitude smaller than other methods. However, this granularity requires models to learn meaningful representations from scratch. Consequently, capturing semantic depth typically requires more complex architectures, such as additional neural layers or larger contextual windows, to compose word-level meaning from character sequences.

4.2 Training Paradigm

This section categorizes matching systems by their training procedures, reflecting their dependency on specific ontology data and their subsequent generalization capabilities. The three categories are: (i) Pre-trained: Systems trained exclusively on corpora external to the input ontologies; (ii) Trained: Systems enploy the input ontologies

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to either train a model from scratch or fine-tune a pre-trained one; (iii) Reference: A subset of the "Trained" category, specifically requiring supervised training on reference alignments.

The **Pre-train** category encompasses systems that employ models trained on external data before the matching process begins. This includes systems that leverage general-purpose pre-trained models, such as BERT, or standard word embeddings. The pre-training itself is often performed on tasks unrelated to ontology matching, with common objectives like masked language modeling or data reconstruction.

The **Trained** category includes systems that require training or fine-tuning using data from the ontologies being aligned, resulting in representations tailored for each matching task. When a system uses a pre-trained model, this process is equivalent to fine-tuning. For example, a system might refine its word embeddings using synonyms extracted directly from the input ontologies. Systems may also be trained on a specific task prior to pre-training on another.

The **Reference** category consists of supervised systems that require an annotated dataset of reference alignments to train their models. This process adjusts the internal parameters of the embedding-based similarity models, such as network weights. These reference alignments, which function as the ground truth, can be created manually by domain experts, generated automatically, or produced by a "teacher" matching system.

4.3 Information Level

This section details the Information Level category, which classifies systems based on the scope of information they aggregate to generate entity embeddings. It distinguishes between four levels: *No Context, Context, Relation,* and *Background Knowledge*. Note that these levels are not entirely independent. Incorporating Background Knowledge from external sources is an approach that can be combined with any other information level. A system that employs information from neighboring entities is classified as Contextual. If it explicitly encodes the type of properties to those entities, it is classified under both the Context and Relation levels.

The **No Context** level includes systems that generate an embedding for an entity using exclusively the information contained within that entity itself. Under this approach, each entity is treated as an isolated object, and information from other entities or their relationships is disregarded. A matcher at this information level generates an entity's final embedding by exclusively using its internal features, such as labels and comments, while disregarding any of its relationships to other entities. For example, a non-contextual embedding for the class `Paper` relies solely on its own internal features, whereas a contextual embedding integrates information from related entities (e.g. $\vec{Paper} = \vec{Article} + \vec{Conference}$).

In contrast, the **Context** level considers not only an entity's internal information but also data from its surroundings and its position in the ontology hierarchy. Systems at this level generate embeddings by aggregating features, such as labels and comments, from neighboring entities (e.g., parent or child classes) to create a richer representation. However, a key distinction at this level is that while the presence of neighboring entities is used, the type of the connecting property is not (e.g., a disjointWith property would be processed in the same manner as a subClassOf property).

The **Relation** level includes systems that explicitly encode the type of properties an entity has with its neighbors when generating embeddings [126]. Unlike at the Context level, a matcher here distinguishes between different kinds of properties, such as inheritance and disjointness. This distinction is important because different properties imply different similarity assumptions. For example, entities in a disjoint property should have lower similarity scores than entities in a parent-child hierarchy. Integrating such relational information guides the model towards embeddings that more accurately reflect these semantic constraints.

The **Background Knowledge** level comprises systems that enrich entity embeddings by incorporating information from sources external to the input ontologies, such as Wikipedia or WordNet [91]. This approach allows a matcher to inject supplementary information beyond what is explicitly stated in the ontology. For

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instance, if an entity lacks a descriptive comment, an external source can provide this missing text, which is then aggregated to generate a more comprehensive embedding. The key distinction from the Context level is the origin of the information: the Context level uses only data already internal to the input ontologies, whereas the Background Knowledge level leverages external data sources, including Wikipedia, Thesaurus, Vocabulary, etc.

4.4 Alignment Cardinality and Complexity

This section categorizes systems based on the alignment complexity they address, distinguishing between simple and complex matching tasks. This classification is important for understanding how the capabilities of different systems vary with the intricacy of the required correspondences. **Simple** alignment systems operate on atomic entities and are designed to produce one-to-one (1:1) correspondences. In this model, each entity from a source ontology is matched to a single, unique entity in the target ontology. In contrast, **Complex** alignment systems handle intricate correspondences that involve complex entities, in other words, those defined by logical combinations of other entities (e.g., using unions, intersections, or restrictions). These systems can produce many-to-many (n:m) alignments and are often necessary for handling constructs such as blank nodes in OWL ontologies.

5 A Survey of Approaches by Information Scope

In this section, the surveyed works are categorized according to the Information Level dimension. This dimension was selected as the primary organizing principle as it directly addresses how ontological information is used to construct embeddings, thus highlighting the core differences between approaches. Within each category, works are further grouped by methodological similarity to illustrate how specific embedding techniques are applied. The embedding models corresponding to each work are summarized in tabular form, using the model categories adapted from [40].

5.1 Approaches Using No Contextual Information

This section reviews systems classified under the No Context category, grouping them by methodological similarity. As defined in Section 4, these systems generate representations for an entity using only its own descriptive information, such as labels and comments, while disregarding data from other entities in the ontology graph (e.g., parent or child classes). The specific models employed by the works in this category are summarized in Table 2.

Word2Vec-Based Representations. Zhang et al. [143] is recognized as the first to apply embeddings to ontology matching. Their study assessed how embeddings could be used to compare the textual information of entities, moving beyond purely lexical metrics. Specifically, techniques such as SVD and Word2Vec were combined with lexical measures to compute similarity. A key limitation of their approach was the absence of a robust method for handling multi-word phrases, such as those commonly found in entity comments. Despite this, the integration of Word2Vec within hybrid similarity pipelines, which also incorporate lexical and hand-crafted rules, was widely adopted in subsequent research.

Following its initial application, Word2Vec was explored across various domains, frequently within a shared architectural framework. A common pattern, observed in systems such as those presented in [64, 98, 134, 137], follows a well-defined pipeline. First, Word2Vec is used to represent the textual information of each entity. For labels containing multiple words, a single embedding is produced via mean aggregation. The resulting embeddings are then compared using cosine similarity, and correspondences are selected from pairs that either exceed a predefined similarity threshold or are chosen through a method such as the stable marriage algorithm to enforce one-to-one correspondences.

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Name	Year	Embedding Model	Embedding Category
Zhang [143]	2014	SCIBOW+DAE	Hybrid/Combined
ERSOM [133]	2015	Word2vec, Glove	Static
Delgado [121]	2017	Stacked Auto-Encoder	Learned Representation (Autoencoder)
DeepAlignment [66]	2018	Learned word embeddings (counter-fitting)	Static (Refined)
OntoEmma [122]	2018	Word2vec, CNN, LSTM, fully connected, Siamese Networks	Hybrid/Combined (Static + Contextualized)
Jurisch [60]	2018	RDF2Vec	KG-Based
Ruiz [59]	2018	Word2Vec	Static
SCBOW+DAE(O) [67]	2018	GLoVe	Static
Gromann [39]	2018	hyperbolic embeddings	KG-Based (Geometric)
Li [73]	2019	RDF2Vec	KG-Based
AnyGraphMatcher [77]	2019	Doc2vec	Static
Dome [46]	2019	StarSpace	Static
Gonçalves [36]	2019	Node2vec, Word2vec	Hybrid/Combined (KG-Based + Static)
Li [71]	2020	Word2vec	Static
DeepFCA [70]	2020	Word2Vec, embedding refinement	Static (Refined)
Hertling [48]	2020	RDF2Vec	KG-Based
Kim [64]	2020	Word2Vec	Static
DESKMatcher [82]	2020	TransE, ConvE	KG-Based
Geng [33]	2020	GloVe	Static
Melis [4]	2020	Fasttext, BERT, SBERT	Hybrid/Combined (Static + Contextualized)
Youn [140]	2020	Word2Vec	Static
REMA [69]	2020	ChatGPT model GPT-4, model agent	Contextualized (LLM)
BERTMap [44]	2021	BERT	Contextualized
Hertling [49]	2021	BERT-base-cased, roberta-base, albert-base-v2	Contextualized
BERTMap [43]	2021	BERT	Contextualized
ALOD2Vec [92]	2021	RDF2vec	KG-Based
Fine-TOM [65]	2021	ALBERT-base-v2	Contextualized
TOM [68]	2021	Siamese BERT, word2vec	Hybrid/Combined (Contextualized + Static)
Li [72]	2021	Word2vec, Glove, kglove	Hybrid/Combined (Static + KG-Based)
Neutel [83]	2021	Word2Vec	Static
Xue [137]	2021	WiSARD is a RAM	Learned Representation
Xue [134]	2022	Word2vec	Static
Portisch [90]	2022	RDF2vec	KG-Based
Rudwan [98]	2022	Word2Vec, RNNLM	Hybrid/Combined (Static + Contextualized)
Mijalcheva [79]	2022	BERT, RDF2Vec	Hybrid/Combined (Contextualized + KG-Based)
A-LIOn [3]	2022	TransR	KG-Based
AMD [127]	2022	Sentence-BERT, TransR	Hybrid/Combined (Contextualized + KG-Based)
KGMatcher+ [29]	2022	BERT	Contextualized
SEBMatcher [37]	2022	ClinicalBERT	Contextualized
WomboCombo [62]	2022	all-MiniLM-L6-v2, ALBERT-base	Contextualized
LEAPME [10]	2022	fastText	Static
WETA [34]	2022	Polyglot, fastText, Word2Vec	Static
Xue [135]	2023	Word2Vec	Static
Qiang [93]	2023	Multi-qa-mpnet-base-dot-v1, upstage/Llama-2-70b-instruct-v2	Contextualized (LLM)
OLaLa [47]	2023	Llama-2 (7b, 13b, 70b, 70b-chat, 70b-instruct-v2), airoboros-l2-70b-2.1	Contextualized (LLM)
Norouzi [85]	2023	ChatGPT	Contextualized (LLM)
Shiraishi [103]	2024	OWL2Vec*, Auto encoder	Satic
Teymurova [115]	2024	Word2vec, OWL2Vec*	Static
Yang [138]	2025	Transbox	Static

Table 2. Embedding models employed by surveyed works classified under the No Context information level category.

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Embedding techniques are also effective for representing entities when combined with complementary methods. For instance, Geng et al. [33] demonstrated this by integrating Word2Vec with Node2Vec. A related approach in [71] highlights the value of domain-specific training, where word embeddings trained on biomedical text were used to match biomedical ontologies. Notably, that study also demonstrated that integrating the proposed embedding technique into existing matching systems yielded performance improvements.

In the multilingual domain, the work in [39] uses Word2Vec and fastText embeddings to generate representations for cross-lingual ontology matching. Their approach computes an intermediate vector based on the cross-similarity between the words contained in two entity labels, a strategy that simultaneously addresses the challenge of OOV terms.

Hybrid Embedding and Lexical Similarity. A prevalent approach in the literature involves constructing hybrid systems that combine embedding techniques with traditional lexical similarity metrics. For example, the work by Fallatah et al. [29] integrates the Levenshtein lexical similarity metric with word embeddings, selecting the maximum value between the two scores to determine the final similarity. Similarly, the approach in [36] first employs lexical metrics to compute an initial similarity matrix, which is subsequently combined with GloVe embeddings trained on PubMed data to produce the final alignment. The DOME matcher [46] also combines lexical and embedding-based methods, but does so sequentially. An initial similarity score is computed using lexical metrics and is then adjusted using Doc2Vec embeddings to produce the final similarity scores. Focusing on the specific domain of property matching, LEAPME [10] employs a distinct hybrid technique that relies on the embedding similarity between labels alongside a set of hand-engineered metrics.

Embedding Refinement and Neural Augmentation. Despite their flexibility, pre-trained embeddings have a notable limitation: they are static, assigning a single, fixed representation to each term. This makes it difficult to handle homonymous entities, whose meaning may vary depending on the context in which they appear. One strategy to address this issue is supervised refinement, a process that contextualises pre-trained embeddings to the specific semantics of the ontologies under analysis. The works in [66, 67] refine a set of pre-trained vectors by contrasting the representations of positively aligned entity pairs with those of negative ones. The DeepFCA matcher [70] implements a related refinement strategy, employing formal concept analysis to generate a lattice of related entities that are then approximated in the embedding space, while negative samples are extracted from disjoint properties.

Another contextualization strategy involves using neural networks to further process or refine existing embeddings. Xue et al. [135] employ Word2Vec embeddings as input features to a neural network with the aim of improving the final similarity computation. A similar approach is adopted in WETA [34], which combines the cosine similarity derived from fastText embeddings with classification scores produced by a neural network. The study in [121] enriches a hand-engineered feature vector with the output of a neural network to compute similarity between entities. In a complementary direction, neural networks can also be used to generate embeddings directly. The ERSOM matcher [133], for example, uses an autoencoder to learn entity representations from a set of input features. Because the autoencoder's training objective is to reconstruct its input from a compressed latent representation, it can be trained in an unsupervised manner, making it particularly suitable for tasks in which reference alignments are unavailable.

Transformer-Based Approaches. With recent advances in NLP, powerful models based on the Transformer architecture, such as BERT, have been widely adopted. BERT is often described as a "contextual" model because it generates a unique representation for a word conditioned on its surrounding text. However, it is important to distinguish this notion of contextuality from the definition used in this survey. Here, entity context refers specifically to information derived from an entity's neighbours within the ontology structure, such as its parent and child classes. Accordingly, systems that apply transformer-based architectures to an entity's textual information in isolation, without relying on this broader ontological context, are classified as "No Context" systems. For instance, the TOM and Fine-TOM matchers [65, 68] use transformer models to generate representations from entity text, which are then compared using cosine similarity.

A prevalent strategy for transformer-based models is fine-tuning on task-specific data. Fine-TOM, for instance, is retrained using known entity correspondences, while the approach in [72] leverages formal concept analysis to derive training data for a sentence transformer. Beyond fine-tuning, models are commonly adapted through specialised downstream classification heads. In ontology matching, a standard approach adds such a head to predict the alignment probability between entity pairs, as demonstrated in [37, 44, 49]. Similarly, the approach in [43] pairs a pre-trained BERT model with a binary classifier, further refining the process by averaging probabilities for multi-label entities and employing inverted indexing to prune the candidate space. Other research explores hybrid integrations: Kardos et al. [62] combine BERT embeddings with traditional lexical metrics, while Neutel et

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al. [83] compare the performance of `[CLS]` token pooling against subword embedding averaging. MDMapper [76] employs language model embeddings within a framework for master data alignment. It introduces a quantitative heuristic for hierarchical similarity by integrating heterogeneous entity and attribute data, alongside property-based navigation and conflict-restricted availability management. This integrated approach ensures efficient, progressive matching, as demonstrated by competitive results on OAEI tracks and significant performance gains over baselines in real-world industrial standards (ETIM and ECLASS).

Graph-Based Embeddings. RDF2Vec represents a foundational graph-based embedding approach in ontology matching. It generates sentence-like sequences through random walks over the ontology graph and subsequently processes these sequences with a Word2Vec model, treating each entity label as a single token to produce unique embeddings. REMA [69] follows a similar pipeline, combining random walks and Word2Vec with cosine similarity for alignment. Beyond direct similarity computation, these embeddings frequently serve as features for binary classification. For example, the system in [60] uses RDF2Vec embeddings to predict how changes in ontology resources affect existing alignments, while AnyGraphMatcher [77] feeds various metrics, such as Euclidean distance and skip-gram probability, into a classifier to predict entity correspondences. To mitigate the scarcity of labelled data, AnyGraphMatcher automatically generates a gold-standard training set from initial similarity scores, thereby enabling supervised learning without manual annotation.

When embeddings for two ontologies are generated independently, their respective vector spaces are often unaligned, rendering standard similarity metrics unreliable. To address this problem, Portisch et al. [90] employ linear transformation techniques to learn a rotation matrix that maps both sets of embeddings into a shared vector space, after which entity similarity can be computed using standard metrics such as Euclidean distance. Other works, such as those by Hertling et al. [48], learn a linear projection in a supervised manner to bring the two embedding spaces into alignment. RDF2Vec has also been combined with other models to extend its capabilities. For instance, DESKMatcher [82] evaluates several embedding models, (including Word2Vec, GloVe, and RDF2Vec) and combines them with lexical similarity metrics. In another example, the work in [79] constructs a hybrid representation by integrating BERT-based embeddings (derived from textual content) with RDF2Vec-based embeddings (generated from the ontology graph structure), evaluating the unified embedding using cosine similarity. RDF2Vec has further been applied to complex matching tasks; for instance, in [1, 2], an RDF2Vec-based strategy is used to generate entity embeddings after a fuzzification method has been applied to the ontology.

Translational Embeddings. Another technique explored in ontology matching is translational embeddings. Like RDF2Vec, these embeddings are derived from the graph structure of ontologies. The method operates by iteratively adjusting embeddings to minimise the distance between a valid triple in the ontology, such as (s, p, o), while simultaneously increasing the distance for corrupted triples in which one element has been altered, for instance (s, p, o'). One of the matching systems that employs translational embeddings is A-LIOn [3], which uses TransR [75], a variant of TransE [14], to generate embeddings in the complex number space. AMD [127] uses BERT to generate initial alignment candidates that are subsequently filtered using TransR embeddings. A primary advantage of these techniques is their unsupervised nature: they do not require reference alignments to learn entity representations. The work in [4] extends translational techniques by modelling entities in hyperbolic space, enabling the embeddings to more effectively capture complex predicates such as inverses and compositional patterns. A related distance-based learning technique is StarSpace [131], which operates on a similar contrastive learning principle, training the model to pull embeddings of similar entities closer together while pushing them apart from randomly selected negative samples, thereby ensuring a clearer separation between entities in the vector space. This technique is also applied in [59], primarily to reduce the candidate search space when working with large ontologies, thus improving computational efficiency and making the alignment task tractable at scale. A related translational technique is applied in TransBox [138], an EL++-closed ontology embedding method designed to represent complex logical expressions in Description Logic EL++ via composition. TransBox addresses limitations of existing methods that focus on atomic entities and struggle with intricate axioms, or

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fail to handle many-to-one, one-to-many, and many-to-many relations effectively. It embeds roles as group transitions of vectors within a box, enabling the modelling of complex relational dynamics and role compositions, and enhances box intersections by extending the representation space to $\mathbb{R}^n$. Experimental results on real-world ontologies demonstrate state-of-the-art performance in predicting complex axioms. Silva et al. [104] proposes CMOM-RS, a novel approach for Complex Multi-Ontology Matching (CMOM) in a 1:n setting, without restrictions on mapping arity or the number of ontologies involved. The method combines an LLM-based strategy [18] that leverages geometric operations over language embeddings to discover complex correspondences, with a lexical strategy based on string similarity. The LLM-based component exploits vector subtraction and cosine similarity to identify the optimal set of target entity names composing a mapping. Evaluated on biomedical ontologies using partial reference alignments and manual expert validation, CMOM-RS improves performance over purely lexical approaches and substantially increases the coverage of complex multi-ontology alignments.

5.2 Approaches Exploiting Contextual Information

This section reviews systems that exploit contextual information derived from neighboring entities to compute entity similarity. Such contextual information typically originates from the class hierarchy (encompassing parent and child classes) and associated properties or instances when available. A variety of approaches have been proposed for aggregating this information, each following a distinct methodology. The section provides a structured overview of these approaches, highlighting how different systems incorporate and process contextual data to improve matching accuracy. The embedding models employed by each work are summarized in Table 3

Contextual Text Enrichment. A first strategy consists of incorporating textual information from an entity's structural context directly into its textual description, thereby enriching its representation before encoding. For example, [56] constructs a sentence that concatenates entity labels with those of two parent classes, which is then encoded using character embeddings and aggregated through an LSTM network; the resulting embedding is passed to a classification head to predict correspondence probability. A similar approach is adopted in [120], where hierarchy labels are incorporated into class entity labels, and domain and range information is appended to property labels; the enriched text is subsequently processed by a sentence transformer, and entity similarity is measured using cosine similarity. This approach additionally applies a reasoner prior to matching to infer new relationships, and targets cross-lingual ontology matching scenarios. Truveta Mapper [6] follows a comparable strategy by generating *smart IDs* for each entity that encode hierarchical information—concretely, a child entity's ID contains the IDs of its ancestor concepts. These smart IDs are fed into the ByT5 model, operating at the character level, to produce final embeddings; the model is further fine-tuned using synonym information extracted from the ontologies. TEXTO [87] similarly combines entity information with label descriptions to compute similarity using language models, a strategy also employed in [119]. In the context of complex ontology matching, [107] proposes an integrated approach that leverages LLMs to generate expressive correspondences. Extending the CANARD framework, the method focuses on matching semantically similar surroundings of instance sub-graphs, guided by user knowledge needs expressed as SPARQL queries. Four architectural modifications are introduced: label embedding similarity (LES), embeddings of SPARQL queries (ESQ), subgraph embeddings (SE), and instance embeddings (IE). Levenshtein similarity is replaced by embedding-based similarity in key pipeline steps, using LLMs to encode textual information for more expressive similarity assessment. Experiments on the OAEI Conference benchmark demonstrate that integrating LLMs significantly enhances performance—particularly in the LES and ESQ configurations, outperforming baseline and state-of-the-art systems in precision and F-measure without requiring reference alignments or fine-tuning.

Embedding-Level Contextual Aggregation. A second strategy aggregates contextual information by encoding neighboring entities separately and combining the resulting embeddings. Sun et al. [109] encode labels from properties, parents, children, and instances using Word2Vec, process them through an LSTM, and subsequently

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Name	Year	Embedding Model	Embedding Category
Ruiz [58]	2018	StarSpace	Static
Schmidt [101]	2018	GloVe	Static
Orji [84]	2018	Word2Vec's	Static
Jurisch [61]	2019	RGCNs	GNN-based
Maldonado [78]	2019	KAEGAN,	Hybrid/Combined
STE + SG [130]	2020	Siamese Transformer-encoders, Siamese Graph Attention networks	Hybrid/Combined (Contextualized + GNN)
Iyer [53]	2020	Universal Sentence Encoder, Siamese Network, Self Attention	Contextualized
Ruiz [57]	2020	StarSpace	Static
Sun [109]	2020	LSTM, Attention	Contextualized
Jiang [56]	2020	Word2Vec (char embeddings), LSTM	Contextualized
DAEOM [129]	2020	BERT, GAT	Hybrid/Combined (Contextualized + GNN)
OntoConnect [16]	2020	fastText	Static
VeeAlign [52]	2020	Universal sentence encoder, Siamese Network, Self Attention	Contextualized
MEDTO [41]	2021	BioBERT, Hyperbolic GNN	Hybrid/Combined (Contextualized + GNN)
ANN-OM [136]	2021	Word2vec	Static
OntoConnect [15]	2021	fastText, LSTM, Recursive Neural Network	Contextualized
VeeAlign [53]	2021	Universal Sentence Encoder, siamese network, Self Attention	Contextualized
ONTOCONNECT [17]	2021	GAT	GNN-based
OTMapOnto [7]	2021	fastText, BioWordVec	Static
LogMapanc -ML [20]	2021	OWL2Vec	KG-Based
An [8]	2022	fastText	Static
Monnin [81]	2022	GCN	GNN-based
Wang [123]	2022	Sentence-BERT, (GAT)	Hybrid/Combined (Contextualized + GNN)
Satti [99]	2022	BERT	Contextualized
CIDER-LM [120]	2022	distiluse-base-multilingual-cased-v2	Contextualized
GraphMatcher [27]	2022	Universal Sentence Encoder, GAT	Hybrid/Combined (Contextualized + GNN)
Tayur [112]	2022	Word2Vec, LSTM, GAN	Hybrid/Combined
Wang [124]	2022	CNN, GCN, Attention,	Contextualized
Khoudja [63]	2022	Auto Encoder, BioWordVec	Hybrid/Combined
Javed [54]	2022	BERT	Contextualized
Truveta [6]	2023	ByT5	Contextualized
Hao [42]	2023	OWL2Vec, GCN, NNLM, RNN	Hybrid/Combined (KG-Based + GNN + Contextualized)
SORBET [38]	2023	SentenceBERT	Contextualized
VersaMatch [32]	2023	Google's Universal Sentence Encoder (USE), Spacy	Contextualized
He [45]	2023	ChatGPT 4	Contextualized (LLM)
LaKERMap [128]	2023	Transformer, TrasnsE	Hybrid/Combined (Contextualized + KG-Based)
Benarab [12]	2023	GloVe, Neural Network	Hybrid/Combined
Teymurova [113]	2023	OWL2Vec, GNN	Hybrid/Combined (KG-Based + GNN)
NLFOA [102]	2023	Sentence-transformers/All-mpnet-base-v2	Contextualized
Peng [87]	2023	GloVe, All-MPNet-base-v2	Hybrid/Combined (Static + Contextualized)
Sousa [106]	2023	all-MiniLM-L6-v2, finish-treebank	Contextualized
Tu [40]	2023	SBERT	Contextualized
Qiang [93]	2024	text-embedding-ada-002, text-embedding-3-large	Contextualized (LLM)
Totoian [118]	2024	gtr-t5-large, BioLord-2023	Contextualized (LLM)
Silva [104]	2024	Sentence-BERT	Contextualized (LLM)
Liu [76]	2024	Sentence Transformer	Contextualized (LLM)
Mei [22]	2025	GNN	GNN/Contextualized (LLM)
Taboada [111]	2025	SBERT	Contextualized (LLM)
Sousa [107]	2025	stella-base, GritLM-7B	Contextualized (LLM)

Table 3. Embedding models employed by surveyed works classified under the Context information level category.

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combine them via a second LSTM augmented with an attention mechanism; final entity similarity is computed using Euclidean distance. A related approach is found in [57, 58], where an inverted index (LexI) is used to generate candidate correspondences, which are then embedded using the StarSpace model. For each candidate mapping, the mean vector of StarSpace word embeddings is concatenated with the embeddings of the entities involved, effectively reducing the search space for large ontologies. LogMap-ML [20] encodes ontologies using OWL2Vec and learns a transformation function via Siamese Networks that aligns the embedding spaces of the source and target ontologies; input pairs consist of entity classes concatenated with their hierarchical paths to the root. OWL2Vec embeddings are also exploited for complex matching tasks, as demonstrated by Teymurova et al. [113]. Dhouib et al. [25, 26] encode entity labels using fastText, average word embeddings to obtain entity representations, and further average these with embeddings from child entities. The specificity of a relation is estimated by

its distance from the cluster center, general relations tend to cluster near the center, while specific ones are more distant. Building on OWL2Vec, OWL2Vec4OA [114, 115] extends OWL2Vec* [19] specifically for ontology alignment. While OWL2Vec* generates embeddings for individual ontologies, OWL2Vec4OA incorporates edge confidence values from seed correspondences (obtained from systems such as LogMap and AML) to bias random walks over interconnected ontologies. Ontology axioms are transformed into RDF triples, projected ontology graphs are merged with weighted edges (1.0 for axioms, confidence values for correspondences), and a biased random walk strategy is applied. This design prioritizes high-confidence edges during walk generation, yielding contextually richer entity embeddings. Evaluations on biomedical ontology alignment tasks from the OAEI Bio-ML track demonstrate significant improvements over OWL2Vec* in MRR and Hits@1 across the OMIM-ORDO, NCIT-DOID, SNOMED-NCIT-N, and SNOMED-NCIT-P datasets. SORBET Matcher [38] encodes entities together with their parent and child entities using Sentence-BERT [96], employing a tree-walking strategy to generate the training data for the matcher.

Similarity-Level Contextual Aggregation. A third strategy computes similarity scores independently for entity information and for contextual data, and subsequently combines them to determine the final correspondence. Xue et al. [136] combine Word2Vec embeddings with lexical measures to compute the embeddings of entities and their parent entities independently; the resulting similarity scores are computed separately and then fed into a neural network to predict the final score. In [101], context similarities are aggregated by blending embedding-based and lexical metrics to obtain initial similarity values, which are subsequently averaged to yield the final score. Similarly, [8] generates entity embeddings using fastText and computes final similarity via Wasserstein distance, where the local context of each entity consists of all triples in which it appears as the subject. Nkisi-Orji et al. [84] propose a range of similarity metrics (including Word2Vec embedding similarity and a context similarity measure assessing the relatedness of parent and child entities) which are used as features in a random forest classifier that predicts entity correspondences. VersaMatch [32] similarly combines multiple similarity metrics as features in a classifier ensemble; some metrics are derived from embeddings provided by the SpaCy library [50], and a blocking strategy is applied to reduce the number of candidate pairs and improve runtime efficiency. Another approach combining embeddings with classifiers is introduced in [103], which proposes a self-matching training method for Inverted-Index Matrix Embedding (InME) and Co-occurrence Matrix Embedding (CoME) models to improve entity subsumption prediction. InME captures global information from annotation axioms (word occurrence locations), while CoME captures local information (word co-occurrences within axioms). Self-matching training enhances a binary classifier's ability to recognize subclass–superclass relationships, improving robustness for isolated entities. Evaluations on the GO, FoodOn, and HeLiS ontologies show that InME with self-matching training outperforms existing ontology embeddings, and that concatenating CoME with OWL2Vec* yields superior performance on the HeLiS ontology. In BERT-INT [54], transformer-generated similarities are combined with structural information via a similarity matrix. PropMatch [106], which focuses on property matching, uses language models to generate embeddings for property-related entities and aggregates the resulting similarities to determine property correspondences.

GNN-Based Approaches. GNNs constitute a particularly expressive class of models for aggregating contextual information in ontology matching. GNNs rely on iterative aggregation steps that combine entity embeddings with those of their neighbors, propagating information through multiple layers to capture context at varying depths. Several GNN variants have been employed in this setting, including message-passing networks, Graph Convolutional Networks (GCNs), and graph attention networks (GATs). Initial entity embeddings can either be learned through a dedicated embedding layer or generated by a language model from the entity's textual description. VeeAlign [52, 53] is a supervised model that initializes entity embeddings using the Universal Sentence Encoder (USE) to encode textual descriptions. Context from neighboring entities is organized into four facets—ancestral entities, child entities, and entities connected via object and datatype properties, each treated as a path; node-level attention aggregates entities within each path, and path-level attention generates a

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unified context embedding. This context embedding is concatenated with the initial entity embedding and passed through a Siamese network to produce the final representation, with cosine similarity used for cross-ontology comparison. Inspired by VeeAlign, GraphMatcher [27] replaces the facet-based aggregation with a GAT, using USE-encoded embeddings as input. Wu et al. [130] similarly employ a GAT but compute initial embeddings using Siamese transformer encoders; final embeddings are compared using cosine similarity, and an automatic threshold estimation module based on Greedy Matching and Stable Marriage algorithms is incorporated. DAEOM [129] adopts an analogous architecture but replaces Siamese transformer encoders with BERT coupled with a pooling layer. In the NLFOA framework [102], the graph structure of ontology entities is converted into a textual representation that encodes hierarchical relationships and connecting properties; this text is then used to generate embeddings for similarity comparison. MultiEA [139] addresses the problem of simultaneously aligning multiple (more than two) KGs in a single pass. The proposed framework embeds entities from all candidate KGs into a shared feature space using a common GNN-based encoder and explores three alignment strategies—mean, anchor, and pairwise—to minimize distances among pre-aligned entities. An inference enhancement technique that incorporates high-order similarities is also introduced. Two new real-world benchmark datasets (DBP-4 and DWY-3) and a novel evaluation metric (M-Hits@K) are proposed to validate the framework. OM-GGAT [22] combines semantic and structural embedding representations using a gated graph attention model. Semantic knowledge and path information are encoded using OWL2Vec, and ontologies are transformed into graph structures; a GAT then assigns attention weights to neighboring nodes for structural feature extraction, and gated networks fuse the semantic and structural embeddings. Experiments on public datasets demonstrate the effectiveness of this approach in improving matching efficiency.

While GNN-based architectures are capable of modeling rich contextual flows, they are typically supervised, requiring reference alignments to train model weights, a significant challenge given the scarcity of labeled ontology matching datasets. To address this limitation, OntoConnect [15] adopts an unsupervised approach that leverages GNNs for embedding generation. A related study [16, 17] uses fastText to generate initial entity embeddings, which are subsequently contextualized through a GNN and trained with a negative sampling strategy. Another variant [15] employs fastText for initial embeddings and an LSTM-based recursive graph network to compute embeddings across different hierarchical levels. Embeddings are averaged and propagated to higher network layers, and the model is trained via a contextual entity prediction objective, enabling cosine similarity computation between source and target embeddings after training. CLOM [11] is an unsupervised cross-lingual ontology matching method that fuses structural and semantic similarity. It addresses the twin limitations of supervised approaches (which suffer from a lack of training data) and current unsupervised approaches that rely solely on lexical similarity. CLOM uses LaBSE, a state-of-the-art multilingual text embedding model, to embed entity descriptions into low-dimensional vectors for semantic similarity computation. These semantic similarities are used to generate initial seed alignments, which in turn inform the learning of structural embeddings (e.g., node2vec, GCN, RGCN, TransE) over a joint graph; a weighted combination of semantic and structural similarities determines the final alignment. Experiments on the MultiFarm dataset across German–English, German–French, and English–French language pairs demonstrate substantial improvements over lexical methods, confirming the complementarity of structural and semantic embeddings for cross-lingual matching.

LLM-Based Contextual Matching. LLMs have emerged as powerful context-aware encoders and reasoning modules for ontology matching. LLMs4OM [35] evaluates the effectiveness of LLMs in ontology matching through a dual-module strategy: a Retrieval-Augmented Generation (RAG) module for candidate selection and an LLM-based matching module for fine-grained alignment. Zero-shot prompting is explored across three ontology representations—class, class-parent, and class-children. Extensive evaluations on 20 ontology matching datasets show that LLMs within LLMs4OM can match or even surpass traditional matching systems, particularly in complex scenarios. The unsupervised, domain-agnostic HybridOM [118] combines vector and string-based similarity through hybrid search. Entity representations are enriched by using LLMs to generate additional descriptions,

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integrating both ontology knowledge and LLM-generated content. The pipeline involves data preprocessing (extracting raw data, verbalizing axioms with DeepOnto, and generating biomedical descriptions with LLaMA3), embedding generation (using gtr-t5-large and BioLord-2023), and ingestion into an OpenSearch vector database; the core matching process combines k-NN and BM25 retrieval with a dual flow-directional search. Evaluations on the Bio-ML 2024 track demonstrate high F1-score and recall, particularly in complex cases. MILA [111] embeds a retrieve-identify-prompt pipeline within a prioritized depth-first search (PDFS) strategy, addressing both the limited performance of LLM-based systems on complex scenarios and their high computational overhead. MILA first identifies high-confidence bidirectional (HCB) correspondences using fast heuristics, reserving LLM queries for borderline cases; candidate retrieval leverages SBERT embeddings for semantic similarity. Evaluated on biomedical, anatomy, and biodiversity tasks from OAEI 2024, MILA achieves the highest F-measure in five of seven unsupervised tracks, matches or outperforms supervised systems, and substantially reduces runtime without requiring domain-specific heuristics or fine-tuning. Agent-OM [93] introduces an agent-based paradigm for ontology matching, consisting of two Siamese LLM agents (a Retrieval Agent and a Matching Agent) with independent planning modules and tools but a shared memory. The Retrieval Agent extracts entity information (metadata, syntactic, lexical, and semantic) and stores it in a hybrid relational-and-vector database.The Matching Agent then performs hybrid database search to identify candidate correspondences, aggregates them via reciprocal rank fusion, validates them through LLM-driven self-checking, and merges bidirectional results. This design mitigates both retrieval cost and LLM hallucination. Evaluations on the OAEI Conference, Anatomy, and MSE tracks show competitive performance on simple matching tasks and significant improvements on complex and few-shot tasks.

5.3 Approaches Encoding Relational Semantics

The context-aware systems described in Section 5.2 aggregate neighbor information into entity embeddings but frequently overlook the specific types of properties connecting entities to their neighbors - information that could substantially improve the quality of the resulting representations. For instance, while disjoint entities tend to be semantically dissimilar, integrating their shared neighborhood into embeddings without distinguishing the disjointness relation can produce the opposite effect, artificially increasing their similarity. To address this limitation, several matching systems explicitly encode relational information when integrating contextual data into entity embeddings. The models employed by the systems in this category are summarized in Table 4. Wang et al. [124] construct a pairwise connectivity graph from the two input ontologies, compute a similarity matrix via a convolutional network, and apply a graph attention network with edge-aware attention to aggregate contextual information into entity embeddings while preserving the semantics of specific edge relations. The approach in [61] creates a unified graph from both source and target ontologies and employs a R-GCN to generate the final embeddings, comparing this strategy against established KG embedding techniques including TransE, TransH, and RDF2Vec. Hao et al. [42] use OWL2Vec to generate initial embeddings and encode structural paths using a two-hidden-layer multi-layer perceptron, representing each subject–predicate–object triple as a paired encoding to produce a unified embedding, a strategy also adopted by Benarab et al. [12]. The MEDTO matcher [41] generates initial entity embeddings using BioBERT and subsequently produces property-aware representations through two complementary methods: a hyperbolic GCN for hierarchical entities and a R-GCN for non-hierarchical ones. The two representations are concatenated and passed to a classification network to predict correspondence probability. Wang et al. [128] introduce a Transformer encoder pre-trained on ontology paths, whose output embeddings are further refined with TransE to produce representations for both classes and properties. Monnin et al. [81] employ a GCN to learn entity embeddings and cluster similar entities, encoding properties as additional edges during a pre-processing step so that they are incorporated into the convolutional aggregation; this method is capable of detecting specific correspondence relation types such as *sameAs*, *broadMatch*, and *related*. Finally, the

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approaches in [78, 112] explore the use of Generative Adversarial Networks (GANs) to generate property-specific embedding representations for the ontology matching task.

Name	Year	Embedding	Embedding Category
Jurisch [61]	2019	RGCNs	GNN-based
Maldonado [78]	2019	(KAEGAN),	Hybrid/Combined
MEDTO [41]	2021	BioBERT, hyperbolic GNN	Hybrid/Combined (Contextualized + GNN)
Monnin [81]	2022	GCN	GNN-based
Tayur [112]	2022	Word2Vec, LSTM, GAN	Hybrid/Combined (Static + Contextualized)
Wang [124]	2022	CNN, GCN, Attention,	GNN-based
Hao [42]	2023	OWL2Vec, GCN, NNLM, RNN	Hybrid/Combined (KG-Based + GNN + Contextualized)
LaKERMap [128]	2023	Transformer, TrasnsE	Hybrid/Combined (Contextualized + KG-Based)
Benarab [12]	2023	GloVe, Neural Network	Hybrid/Combined
NLFOA [102]	2023	Sentence-Transformers/all-mpnet-base-v2	Contextualized
Yang [139]	2024	GNN	GNN

Table 4. Embedding models employed by surveyed works classified under the Relational Semantics information level category.

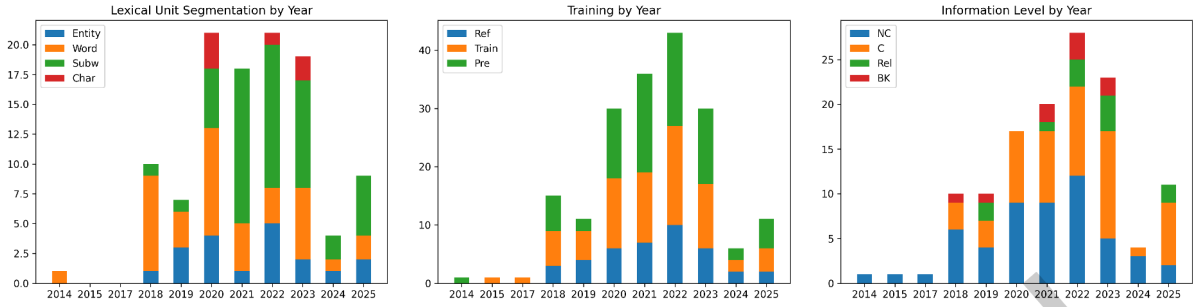
5.4 Approaches Leveraging Background Knowledge

Ontologies frequently lack exhaustive information about all their entities; some entities may have only a label and no accompanying definition or comment. In such cases, the available information may be insufficient for constructing adequately representative embeddings. To address this limitation, several matchers inject background knowledge from external sources into the embedding generation process. The authors in [122] use Word2Vec embeddings to encode entity textual information, which is subsequently aggregated through an LSTM. Four views are employed (names, synonyms, definitions, and Wikipedia context) to construct entity embeddings that are passed to a neural classifier for correspondence prediction. Similarly, Wang [123] relies on Wikipedia as background knowledge through three complementary views: a semantic view generated via an LSTM, a structural view aggregated using a graph attention network, and a definition view embedded using Sentence-BERT; the combined representation is then used to compute cosine similarity. Li et al. [73] adopt a further multi-view strategy, composing final embeddings from lexical, structural, and resource views, where the resource view links source and target entities to an external reference ontology. Satti et al. [99] use entity names and types as the basis for embedding composition, querying UMLS with suffixes extracted from these texts to retrieve supplementary information, which is subsequently encoded by a Transformer model. In OTMapOnto [7], fastText and BioWordVec embeddings are averaged to produce entity representations, enriched with synset information from WordNet, and compared using the Wasserstein distance. The ALOD2Vec matcher [92] queries an external RDF2Vec-based vector dataset using entity lexical information and computes final similarity via cosine similarity. Finally, Khoudja et al. [63] feed a set of similarity metrics (including similarity to reference ontologies) as background knowledge, into an autoencoder to generate the final entity embeddings. The models employed by the systems in this category are summarized in Table 5.

Name	Year	Embedding	Embedding Category
OntoEmma [122]	2018	word2vec, CNN, LSTM, fully connected, siamese networks	Hybrid/Combined (Static + Contextualized)
Li [73]	2019	TransE, ConvE	KG-Based
ALOD2Vec [92]	2021	RDF2vec	KG-Based
OTMapOnto [7]	2021	fastText, BioWordVec	Static
Wang [123]	2022	Sentence-BERT, (GAT)	Hybrid/Combined (Contextualized + GNN)
Satti [99]	2022	BERT	Contextualized
Khoudja [63]	2022	Auto Encoder, BioWordVec	Hybrid/Combined (Learned + Static)
VersaMatch [32]	2023	Google's Universal Sentence Encoder (USE), spacy	Contextualized
Benarab [12]	2023	GloVe, Neural Network	Static + GNN

Table 5. Embedding models employed by surveyed works classified under the Background Knowledge information level category.

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(a) (b) (c)

Fig. 8. Variations in each group over time. (a) Lexical unit segmentation usage over time. (b) Training approaches usage over time. (c) Information level usage over time.

5.5 Comparison

Each paper is categorized according to the proposed categorization scheme. The results for simple alignments are presented in Table 7 under the **Non-context** category, and Table 8 contains the results for works integrating contextual information. The results for complex matching are reported in Table 9. Several trends emerge from the **Lexical Unit Segmentation** dimension, illustrated in Figure 8(a). Embeddings were first applied to ontology matching in 2014, and between 2015 and 2017 hand-engineered embeddings predominated. Thereafter, word embeddings became the most widely used approach, accompanied by a rise in entity-level embeddings (particularly those based on RDF2Vec). The subsequent development of Transformer models drove subword embeddings to become the dominant segmentation method, while the use of word embeddings gradually declined. The growing adoption of GNNs has renewed interest in entity-level and character-level embeddings, and the emergence of LLMs has begun to replace conventional full-embedding approaches with LLM matchers in recent years. In the **Training** dimension (Figure 8(b)), a consistent increase is observed in both the use of pre-trained models and the use of reference alignments for embedding generation. In the **Information Level** dimension (Figure 8(c)), a growing proportion of works employ contextual information to generate entity embeddings, a trend accompanied by a corresponding decline in purely non-contextual approaches. Alongside this shift, increasing attention is being directed toward encoding property-specific data into embeddings and toward integrating background knowledge.

The distribution of embedding techniques across the surveyed literature is summarized in Figure 9. In the Lexical Unit Segmentation category, subword (45%) and word (36%) segmentation predominate, largely driven by the prevalence of pre-trained transformer models and the ease of integrating word embeddings. Entity-level segmentation accounts for 18%, limited by its poor domain generalization and the scarcity of pre-trained models. Character segmentation is the least utilized (6%), as the high complexity of training such models hinders their adoption. Regarding Training, both pre-trained (70%) and ontology-specific trained models (68%) are dominant. Reference alignment usage remains significant at 38%, despite the risk to generalization given that such alignments are not universally available. Within the Information Level category, usage is evenly split between no-context and context-aware embeddings (50% each). However, specific techniques like property encoding (12%) and background knowledge integration (9%) remain underexplored. Finally, for Alignment Type, 93% of works focus on simple matching, with only 7% addressing complex alignments. Given that simple alignments cannot resolve all ontological heterogeneities, increasing the focus on complex matching represents a critical opportunity for advancing data integration in this field.

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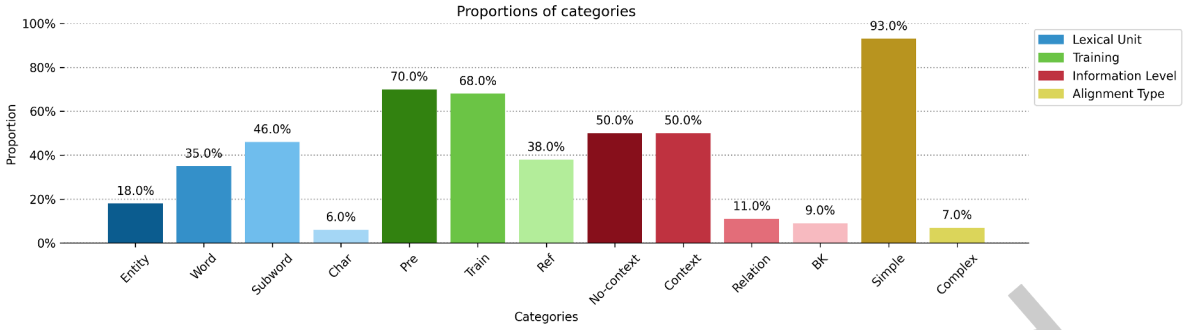


Fig. 9. Distribution of the proposed classification dimensions across all surveyed papers.

To analyze the distribution of category combinations, all possible multi-dimensional combinations were enumerated and their frequencies normalized by the total number of papers; the results are reported in Table 6. The most prevalent combination involves subword and word lexical segmentation, reflecting the dominance of the most common models used in embedding generations. Character-level segmentation remains underexplored, appearing in only a few papers across all combinations. Within the training category, entity-level segmentation is rarely used in conjunction with pre-training and has few associated applications of contextual embeddings. At the information level, only four papers in the top 15 combinations apply property-aware embeddings, and no background knowledge integration.

Combination	Count	Proportion
Subw-Pre-C	26	0.25
Subw-Pre-NC	22	0.21
Subw-Train-C	15	0.14
Word-Pre-NC	14	0.13
Word-Train-C	13	0.12
Entity-Train-NC	11	0.10
Word-Train-NC	11	0.10
Word-Pre-C	10	0.10
Subw-Ref-C	10	0.10
Subw-Train-NC	10	0.10
Entity-Ref-NC	7	0.07
Word-Ref-C	7	0.07
Entity-Train-C	6	0.06
Entity-Ref-C	5	0.05
Entity-Train-Rel	5	0.05

Table 6. Frequency and relative proportion of 15 most common multi-dimensional category combinations across the surveyed works.

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6 Open Challenges and Research Gaps

6.1 Trends in Embedding-Based Ontology Matching

The analysis conducted in this survey reveals several challenges requiring further attention.

Generalization problem. A recurrent limitation associated with entity-level segmentation in ontology matching is its restricted capacity to generalize across domains. The absence of a shared vocabulary needs retraining the model for each previously unseen ontology, thereby hindering the cumulative acquisition of knowledge across matching tasks. For example, entity embeddings learned in the biomedical domain are unlikely to contain the same terminology as those required in the conference domain. Furthermore, even in cases where an entity label

appears identically across domains, its semantic interpretation may differ substantially from that encoded during training, giving rise to ambiguity. Without the capacity to exploit patterns in textual information, generalizing a matcher remains challenging even within a single domain. As the number of unique embeddings to be learned grows, sparsity issues emerge in the embedding space, complicating knowledge reuse. Additionally, without analyzing the textual information for each entity, it becomes impossible to eliminate noisy information or to leverage commonalities in knowledge, such as comments and descriptions.

Lack of shared vocabulary. Another issue that arises from this level of segmentation is the lack of shared vocabulary, which prevents embeddings from different ontologies from relating to each other. Due to the random initialization of neural models, the same entity can have different embeddings in two distinct ontologies when learned through these methods, as they are encoded in separate embedding spaces. The work in [90] illustrates this problem in embeddings generated by RDF2Vec using two identical copies of the same graph generated from synthetic data. To address this, the authors propose learning a rotation matrix from anchor correspondences to transform one embedding space into the other. They also demonstrate that introducing heterogeneity, by randomly removing triples from one of the graphs, significantly reduces matching performance. A similar issue was observed in LogMap-ML [20], in which OWL2Vec embeddings were used for matching. The independence problem, caused by the generation of separate embedding spaces during independent OWL2Vec training, was resolved by feeding the embeddings into a siamese network. It was trained using anchor correspondences to learn a transformation function that aligns the two embedding spaces.

Structural divergence. Embedding methods that rely exclusively on ontology structure, without incorporating the textual information associated with each entity, are inherently sensitive to structural differences between the ontologies being aligned. Consequently, such methods tend to exhibit reduced performance when applied to ontology pairs with substantially divergent structural organizations. This limitation is evidenced in Hertling et al. [**?**], where RDF2Vec is evaluated across ontologies from different domains, yielding strong results when the compared ontologies share similar structural patterns but markedly lower performance when their structures differ. Although RDF2Vec effectively encodes the structural relationships among entities within an ontology graph, structural proximity does not necessarily entail semantic equivalence - the primary objective of alignment. For instance, entities such as "Author" and "Book" may co-occur frequently within the same ontological neighbourhood due to their relational proximity, causing their embeddings to be placed in close proximity in the vector space. Despite exhibiting high cosine similarity, these entities are semantically distinct, and a matcher relying solely on such structural embeddings risks producing erroneous correspondences between them.

Need for training alignments. A significant portion of the reviewed systems (48%) relies on reference alignments for training. These alignments are typically generated by domain experts or automatic systems and establish equivalence relations between entities across ontologies. As no large alignment dataset is currently available, many systems resort to the reference alignments provided by the OAEI tracks for model training. However, since these alignments are also used for evaluation, systems trained on such data may suffer from generalisation issues. Alternative approaches treat other matching systems as teachers, using their alignment outputs as training data, or adopt alignment hypotheses such as exact label matching as a gold standard. These strategies assume that embedding models can extrapolate knowledge and identify additional correspondences, thereby increasing the number of available alignments. However, when such systems are also evaluated in competitive settings, alignment biases may arise, further resulting in generalization problems. To address these limitations, unsupervised training methods have been proposed, embedding ontologies into a shared vector space and inferring alignments from entity similarity. However, these approaches remain under development and require further investigation to assess their effectiveness.

Coverage of properties. Few works explore the integration of property-specific information into the embedding generation process for similarity computation. Certain properties, such as disjointness, indicate a negative effect on similarity, and merely including the object values they represent can lead to misleading results, causing

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disjoint entities to appear more similar than they are. Integrating property-specific information into embedding generation can therefore enhance the accuracy of alignment systems. For instance, incorporating domain and range information into property embeddings, or explicitly encoding disjointness and other constraints, can provide a richer context for entity representations.

Background knowledge. Few systems investigate the integration of background knowledge as supplementary information for generating embeddings. Integrating such external knowledge can offer additional context and information for entities that lack sufficient resources. For example, leveraging external knowledge sources like Wikipedia or WordNet can enrich entities with textual descriptions or semantic annotations, improving the quality of the generated embeddings. However, this integration is not without drawbacks, as the incorporation of external data may introduce noise into the embedding space and increase the computational overhead of the alignment process. Consequently, further research is needed to examine the trade-offs between accuracy and efficiency in matching systems that adopt such strategies based on the integration of external information during embedding generation.

Complex matching. Despite the importance of complex correspondences in real-world ontology alignment, only a limited number of approaches have explored the application of embeddings to complex matching tasks. In such tasks, the aggregation of multiple constituent entities into a unified representation of a complex entity introduces additional difficulty in defining the boundaries and scope of meaningful contextual information. Specifically, a matching system must determine the extent to which the contextual information associated with each constituent entity contributes to the overall context of the composed complex entity. When complex entities and their potential contexts are modeled as subgraphs, the number of candidate combinations to be considered grows exponentially, resulting in worst-case time complexity that poses a significant scalability barrier in complex matching scenarios.

6.2 Answering the Research Questions

This survey presents a systematic analysis of 103 embedding-based ontology matching approaches, providing evidence-based answers to the research questions formulated in Section 1. In the following, the each research question is addressed in turn, synthesizing the key findings across the four proposed classification dimensions (lexical segmentation, training paradigm, information scope, and alignment complexity) with reference to the surveyed literature.

RQ1: How are embeddings used to represent ontological entities for matching? Embeddings are one of the main semantic similarity measures of current matching systems, and their representational capacity varies based on three critical design choices revealed through this survey analysis:

- the choice of **lexical segmentation strategy** has a strong impact on the generalization capability of the embeddings in a matcher: the survey shows 46% of approaches employ subword segmentation (primarily via transformer-family models), 35% use word-level segmentation, 18% adopt entity-level representations (e.g., RDF2Vec), and only 6% utilize character-level approaches. Entity-level embeddings face generalization limitations, as demonstrated in [90], because most of the techniques used in creating those embeddings often produce them independently for source and target ontologies that occupy different vector spaces, requiring alignment techniques like rotation matrices to make them reflect the similarity measure required.
- the **information level** determines the amount of information considered for similarity computation: half of surveyed systems generate embeddings considering only entity-related features (labels, comments), while the other half incorporate contextual information from neighboring entities. Crucially, only 12% explicitly encode relational semantics (e.g., distinguishing `subClassOf` from `disjointWith`). Not being able to encode properties in the embeddings can cause erroneous matches (e.g., placing semantically disjoint entities close in

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embedding space). Then, background knowledge integration remains underutilized (9% of systems), though works like [123] demonstrate its value for entities lacking descriptive text.
- considering **training-related properties** and the **types of correspondences** produced by matching systems, a notable pattern emerges: while 70% of systems employ pre-trained embedding models, 68% additionally fine-tune these models on ontology-specific data in order to adapt representations to domain semantics. This trend suggests that general-purpose pre-trained models are insufficient for capturing domain-specific similarities in isolation, necessitating some degree of task-specific adaptation. Additionally, 38% of the reviewed works require reference alignments during training. The reliance on such alignments introduces a practical limitation, as curated alignment data is scarce.

RQ2: What types of embedding strategies have been applied to ontology matching, and how do they differ? The analysis in this survey reveals the usage of multiple types of embedding strategies, but most of them follow some trends. The dominant paradigm has evolved from static word embeddings (2015–2018) toward transformer-based contextual embeddings (2019–present), but with critical variations:

- **Lexical/Semantic Embeddings**: These approaches use word embeddings (e.g., Word2Vec, GloVe) or contextual embeddings (e.g., BERT) to capture semantic information from entity labels and textual descriptions. They emphasize language understanding but often overlook ontology structure.
- **Graph-Based Embeddings**: These methods represent ontologies as graphs and use algorithms like RDF2Vec, TransE, or GCNs to learn embeddings that encode structural information. They can better capture hierarchical and relational patterns, but may struggle with sparse or inconsistent graphs.
- **Hybrid Approaches**: Combining lexical and structural features, hybrid models aim to leverage the strengths of both. Some methods concatenate multiple embeddings, while others use attention or fusion mechanisms. These tend to perform better in varied scenarios but may be more complex and computationally demanding.

The primary differences lie in the types of input data they use (text vs. structure), the embedding architectures (e.g., static vs. dynamic), and the extent to which they generalize across domains.

RQ3: How do existing methods handle the complexity and heterogeneity of real-world ontologies? Current methods address heterogeneity through four evidence-based strategies, but with significant limitations revealed in this work:

- **Multi-modal embeddings** shows promise but remains under-engineered: while most works combine lexical and structural features, only 9% incorporate background knowledge. Also, the mechanisms of embedding combination are often simplistic (e.g., averaging similarities), which can introduce noise in the similarity computation. Some works demonstrate that property-aware graph networks can better handle medical ontologies' deep hierarchies, but such integration is difficult to implement, and not all works explored them.
- **Context-aware adaptation** addresses lexical heterogeneity effectively: approaches incorporating hierarchical context (e.g., parent/child labels) can achieve higher F-measure since aggregating this information can improve the similarity computation used in the matching process however, as can be seen in Table 6 from the total distribution of all classification combinations of embedding used in the reviewed works, the relation aware embeddings appears in only 0.05% in total. This reveals that most context aggregation methods treat all neighbors equally, ignoring that `disjointWith` relations, as described in RQ1 discussion, should *decrease* similarity while `subClassOf` should increase it.
- **Focused on simple correspondences** Most of the current approaches focuses in finding only simple correspondences. Only 7% of surveyed works propose approaches that can find complex correspondences. Most of the real-world ontologies require those types of correspondences to be aligned. Since only a few works propose applications of embeddings in such cases, this demonstrates that the proposed approaches still have difficulties in dealing with real-world ontology heterogeneities.

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Along with those considerations, only one work explored the generation of embeddings aggregating both relational and background knowledge in the information level category. This analysis demonstrates that most of the approaches are still focusing on one type of heterogeneity and still can not handle all types of heterogeneities simultaneously.

RQ4: What are the limitations of current embedding-based approaches, and what directions should future research take? This synthesis identifies four interconnected limitations that form a research agenda grounded in empirical evidence:

- **The supervision bottleneck**: 38% of approaches require reference alignments for training, creating a circular dependency where matching quality depends on existing alignments. This severely limits applicability to novel domains. Unsupervised approaches exist, but underperform supervised ones. Future work must develop *weakly supervised* paradigms that leverage minimal human input (e.g., fewer seed alignments) combined with self-training on promising directions that also extend to complex matching scenarios.
- **The structural semantics gap**: Current embeddings poorly represent logical constraints (e.g., disjointness, cardinality restrictions, intersections). This analysis shows only 7% of systems handle complex correspondences involving logical constructors—despite most of the real-world ontology alignments requiring such expressions [116]. Future embeddings must move beyond node/edge representations to encode complex entities (constructed with logical predicates such as union or intersection), potentially through neuro-symbolic integration or geometric embeddings in constraint-satisfying spaces (e.g., hyperbolic manifolds for hierarchy preservation), also dealing with the embedding shared vector space problem.
- **The explainability deficit**: Embedding-based matches provide similarity scores without justification, hindering expert validation. While the current approaches are searching for improved f-measure scores, incorporating explainability mechanisms is still in the second plan. Given that ontology matching can occur in high-stakes domains (healthcare, finance), future work must develop *interpretable similarity metrics* that surface evidence for matches. One possible direction is through attention visualization over ontology subgraphs or counterfactual explanations.
- **The scalability-complexity trade-off**: Hybrid approaches achieve the highest accuracy but scale poorly to large ontologies (>100K entities). To achieve the top performance, the approaches often sacrifice recall for efficiency by filtering some entities. Future research should explore strategies that allow for efficient embedding generation without sacrificing performance by applying better space reduction techniques that filter entities with low similarity probability.

These limitations aren't isolated technical challenges but interconnected symptoms of a deeper issue: embedding approaches have prioritized relational representation capacity over *similarity-aware* embeddings. Future breakthroughs will come from embeddings explicitly optimized for the alignment task's unique requirements for semantic equivalence detection under heterogeneity, with minimal supervision, and maximal interpretability.

7 Future Research Directions

The study of entity-level segmentation and information-level aggregation shows that embedding models for ontology matching benefit from aggregating as much information as possible. Given the typical structure of ontology graphs, comprising nodes with textual features and logical properties, a model capable of representing this information would likely combine language models and GNNs to embed both specific properties and logical constraints. However, jointly training these two model types is computationally expensive, as high-performing language models typically comprise a large number of layers that incur significant processing overhead, while graph neural networks must operate over large-scale graphs containing millions of triples. Therefore, more efficient methods for training these models in tandem need to be developed to enable the computation of efficient data batches and gradient updates, driving the learning of parameters for both GNNs and language models.

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With the right combination of these approaches and optimal lexical segmentation, a matcher can achieve better representation capacity, generalize more effectively across domains, and accumulate knowledge, ultimately resulting in improved alignment performance.

Even when a suitable model exists for capturing the features necessary to generate embeddings for similarity tasks, such as identifying equivalent entities in ontology matching, the availability of adequate training data remains a critical requirement. Neural models generally require large volumes of data to learn their objectives effectively, and in the context of ontology matching, such data takes the form of reference alignments between ontologies. These alignments are typically produced either manually by domain experts or with the assistance of automatic matching systems, both of which render the data generation process time-consuming and resource-intensive. Moreover, matching systems must not only identify correct correspondences but also reliably reject incorrect ones, resulting in a highly imbalanced training set in which true correspondences constitute only a small fraction of all possible entity pairs. This issue becomes even more pronounced in complex matching scenarios, where the number of candidate subgraph combinations grows exponentially. In this context, the construction of large-scale, openly available alignment datasets would substantially benefit supervised models by providing the breadth of examples needed to learn robust similarity patterns and, consequently, produce more accurate embeddings. As a complementary direction, the refinement of unsupervised techniques that rely on pretext tasks represents a viable alternative for reducing data dependency while maintaining competitive performance in ontology matching.

Recent advancements in NLP have led to the development of LLMs, which possess billions of parameters and demonstrate significant abilities in language comprehension and generation. Given their success in various semantic tasks, LLMs have the potential to enhance performance in Ontology Matching tasks. For instance, He et al. [45] apply LLMs in conjunction with BERTMap, prompting the model to assess the semantic equivalence of entity pairs and using its responses to establish the final correspondences. Another work [93] leverages LLMs by retrieving information from ontologies and constructing prompts to assess entity similarities. Similarly, OLaLa [47] uses LLMs to compute core similarity metrics between entities, with different prompt sets tested to evaluate their impact on performance. In [85], the authors propose using ChatGPT [86] for ontology alignment by querying the language model with a template to assess the similarity of ontology entities. Those trends demonstrate that LLMs will be even more present in the scenario of ontology matching in future works.

Another promising direction for future research lies in the further investigation of the underexplored areas identified by the proposed classification scheme. In particular, the development of more robust techniques for encoding relational properties within embeddings, as well as the systematic integration of external data sources, represent valuable yet insufficiently addressed capabilities in current matching systems. With respect to alignment complexity, complex ontology matching remains a largely underexplored frontier, especially concerning the application of embedding-based methods. Future research in this area should prioritize three interconnected challenges: the construction of dedicated benchmark datasets, the design of principled methods for subgraph selection, and the advancement of techniques for embedding aggregation in the context of complex correspondences.

8 Conclusion

This survey presents a comprehensive analysis of the role of embedding-based approaches in ontology matching. The methodology is described in detail, encompassing the paper selection criteria, the databases queried, and the manual post-filtering process applied, resulting in a curated corpus of relevant works. Building upon a prior classification scheme, the framework was extended by introducing a new dimension distinguishing between simple and complex matching. This addition enables a more nuanced categorization of existing approaches, facilitating the grouping of works according to how embedding techniques are employed within their respective

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matching processes. The analysis uncovers a set of ongoing challenges and limitations that must be addressed in order to advance matcher performance. Notably, while entity-level segmentation holds potential for improving cross-domain generalization, the number of works tackling complex matching tasks remains limited, representing a significant gap in the current literature.

Several limitations of existing works were identified, and directions for future research on the use of embeddings in ontology matching were outlined. One promising avenue is the integration of language models with GNNs, which would allow systems to jointly leverage textual data and relational information encoded in ontology graphs. Another critical direction concerns the development of large, openly available alignment datasets, which would enable supervised models to more effectively learn the semantic representations required for ontology entity matching. Taken together, these findings underscore the utility of the proposed classification scheme as a guiding framework for the analysis of embedding-based approaches in ontology matching. By organizing existing systems along well-defined dimensions, the framework allows researchers to more readily compare methodological choices and identify opportunities for the development of more robust and effective matching techniques.

In conclusion, the findings presented in this survey highlight the importance of embedding-based approaches in ontology matching and emphasize the need for further research in this area. Several directions of future work include: the development of contextual and explainable embeddings that coherently integrate semantic, structural, and justification mechanisms; the advancement of cross-lingual and cross-domain embeddings capable of supporting more diverse and heterogeneous ontology matching scenarios; the design of interactive, human-in-the-loop systems that systematically incorporate expert feedback into the embedding refinement process; and the construction of scalable models able to handle large-scale ontologies efficiently without compromising matching accuracy.

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A.1 Classification of the reviewed works in the proposed categorization

A.2 Analysis of the works using embeddings that participated in OAEI

The results of the matchers using embeddings that participated in the OAEI campaign from 2021 to 2024 are present in Figure 10. Among the matchers that participated, KGMatcher was one of the most successful approaches considering all tracks in 2021. It has a high F-measure in the tracks BioDiv, LargeBio, and Phenotype. Considering its classification in this survey, it has subword lexical unit segmentation, and it is pre-trained and trained with data for each ontology. Despite its performance, KGMatcher uses no-context embeddings without the use of background knowledge. ALOD2Vec is the second one, having a high F-measure in the tracks KnowledgeGraph and LargeBio. ALOD2Vec uses entity-level segmentation and is pre-trained only, not requiring extra training. It also includes background knowledge. Other notable systems include OLaLa, which performed well in the BioDiv track in 2023 and applies subword segmentation with pre-training only. In the Anatomy track, recent systems like SEBMatcher (2022), MDMapper (2024), and SORBETMatcher (2023) also show competitive results. These methods typically use subword segmentation and both training and pre-training, though they do not leverage background knowledge. In the Conference track, GraphMatcher achieved high performance in both 2023 and 2021. It uses subword segmentation, is trained and pre-trained, and relies on reference alignment, but does not incorporate background knowledge. For the LargeBio track, OTMapOnto (2021) stands out with a combination of word and subword segmentation, pre-training, and the use of background knowledge. Lastly, CIDER-LM, the top performer in MultiFarm 2022, also uses subword segmentation, is trained and pre-trained, and is based on reference alignments, though it does not include relational reasoning or background knowledge.

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Name	Entity	Word	Subw	Char	Ref.	Trained	Pre-train	BK
Zhang [143]		X					X	
ERSOM [133]						X		
Delgado [121]						X		
DeepAlignment [66]		X				X	X	
OntoEmma [122]		X					X	X
Jurisch [60]	X				X	X		
Ruiz [59]		X				X		
SCBOW+DAE(O) [67]		X			X	X	X	
Gromann [39]		X	X				X	
Li [73]	X				X	X		X
AnyGraphMatcher [77]		X			X	X		
Dome [46]		X				X		
Gonçalves [36]		X					X	
Li [71]		X		X			X	
DeepFCA [70]		X		X		X	X	
Hertling [48]	X				X	X		
Kim [64]		X					X	
DESKMatcher [82]	X	X					X	
Geng [33]		X				X		
Melis [4]	X					X		
Youn [140]		X	X				X	
REMA [69]	X					X		
BERTMap [44]			X			X	X	
Hertling [49]			X		X	X	X	
BERTMap [43]			X			X	X	
ALOD2Vec [92]	X						X	X
Fine-TOM [65]			X		X	X	X	
TOM [68]			X				X	
Li [72]			X		X	X	X	
Neutel [83]			X			X	X	
Xue [137]		X					X	
Xue [134]		X				X		
Portisch [90]	X				X	X		
Rudwan [98]		X				X	X	
Mijalcheva [79]	X		X		X	X	X	
A-LIOn [3]	X				X	X		
AMD [127]			X			X	X	
KGMatcher+ [29]			X			X	X	
SEBMatcher [37]			X		X	X	X	
WomboCombo [62]			X				X	
LEAPME [10]		X			X	X	X	
WETA [34]			X				X	
Xue [135]		X			X	X		
Qiang [93]			X				X	
OLaLa [47]			X				X	
Norouzi [85]			X				X	
Taboada [111]			X				X	
MDMapper [76]			X				X	

Table 7. Works that the usage of embeddings falls into the **Non-Context** category.

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Name	Entity	Word	Subw	Char	Ref.	Trained	Pre-train	Rel	BK
Ruiz [58]		X				X			
Schmidt [101]		X					X		
Orji [84]		X			X	X	X		
Jurisch [61]	X				X	X		X	
Maldonado [78]	X				X	X		X	
STE + SG [130]		X					X		
Iyer [53]			X		X	X	X		
Ruiz [57]		X				X			
Sun [109]		X			X	X	X		
Jiang [56]				X	X	X	X		
DAEOM [129]			X		X	X	X		
OntoConnect [16]			X			X	X		
VeeAlign [52]			X		X	X	X		
MEDTO [41]			X		X	X	X	X	
ANN-OM [136]		X			X	X	X		
OntoConnect [15]			X			X	X		
VeeAlign [53]			X		X	X	X		
ONTOCONNECT [17]			X			X	X		
OTMapOnto [7]		X	X				X		X
LogMapanc -ML [20]		X			X	X	X		
An [8]			X				X		
Monnin [81]	X				X	X		X	
Wang [123]			X		X	X	X		X
Satti [99]			X				X		X
CIDER-LM [120]			X		X	X	X		
GraphMatcher [27]			X		X	X	X		
Tayur [112]				X	X	X	X	X	
Wang [124]						X	X	X	
Khoudja [63]						X			X
Javed [54]			X				X		
Truveta [6]				X	X	X	X		
Hao [42]		X			X	X		X	
SORBET [38]			X		X	X	X		
VersaMatch [32]			X			X	X		X
He [45]				X			X		
LaKERMap [128]		X				X	X	X	
Benarab [12]	X	X				X	X	X	X
Teymurova [113]		X			X	X			
NLFOA [102]			X		X	X	X	X	
Peng [87]			X				X		
Sousa [106]		X	X				X		
Tu [40]			X			X	X		
Shiraishi [103]		X				X			
Totoian [118]			X				X		
Giglou [35]			X				X		
Chen [22]		X			X	X			
Teymurova [114]		X				X			
Banerjee [11]	X				X	X		X	
Qiang [93]			X				X	X	
Yang [139]	X				X	X			

Table 8. Works in which the employed embeddings fall into the **Context** category.

Name	Entity	Word	Subw	Char	Ref.	Trained	Pre-train	NC	C	Rel	BK
Dhouib [26]			X				X		X		
Dhouib [25]			X				X		X		
Akremi [1]	X					X		X			
Akremi [2]	X					X		X			
MC Silva [104]			X				X	X			
Sousa [107]			X				X		X		
Yang [138]	X				X	X		X			

Table 9. Works that have any usage of embeddings in the Complex Matching task.

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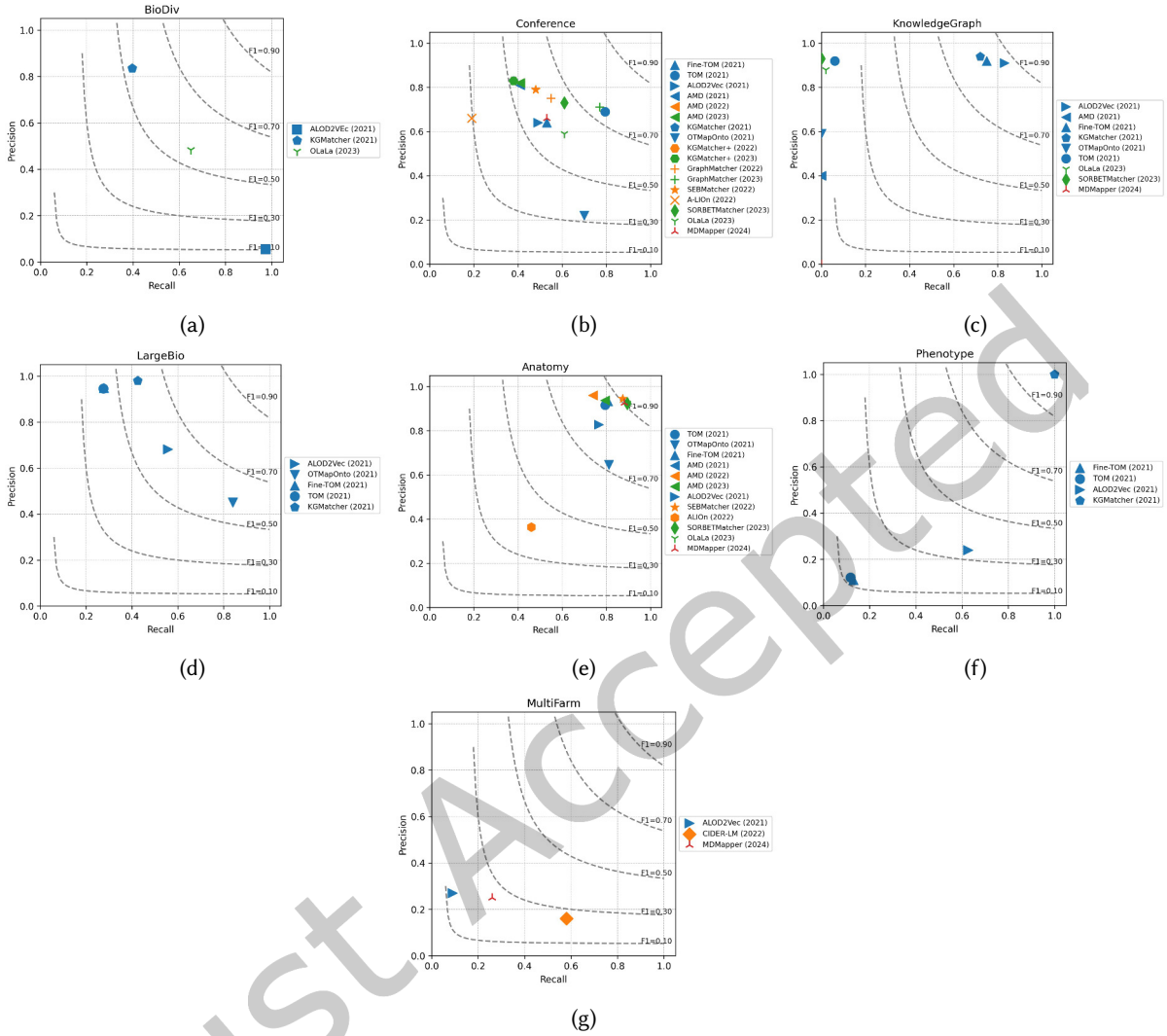


Fig. 10. Performance of embedding matchers in OAEI tracks.

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